# Signals and Systems

## Lecture 5: Z Transform

## Relations between CT and DT

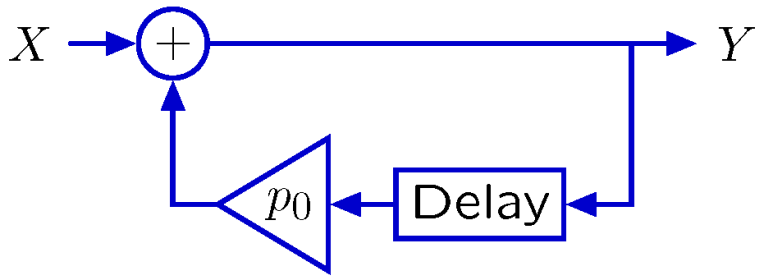
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Zhejiang University

03/11/2025

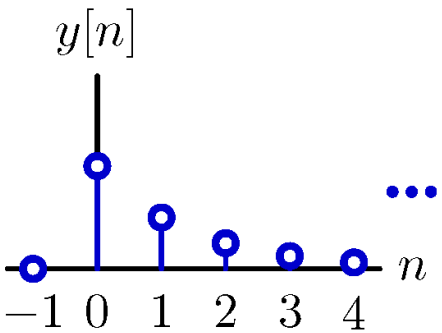
Partly adapted from the materials provided on the MIT OpenCourseWare

# Review: Geometric Growth: Poles

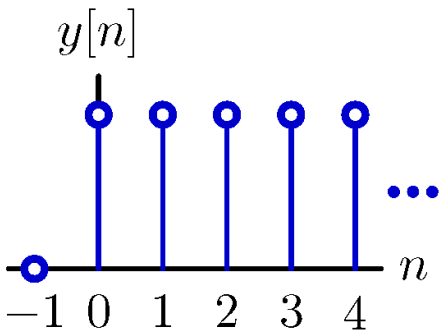
These unit-sample responses can be characterized by a single number — the **pole** — which is the base of the geometric sequence.



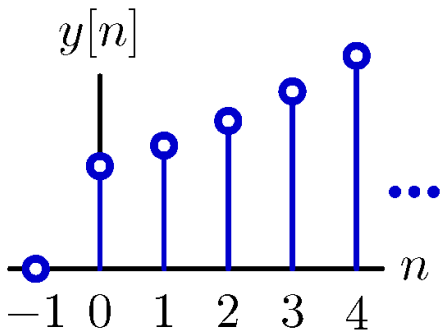

$$y[n] = \begin{cases} p_0^n, & \text{if } n >= 0; \\ 0, & \text{otherwise.} \end{cases}$$




$p_0 = 0.5$



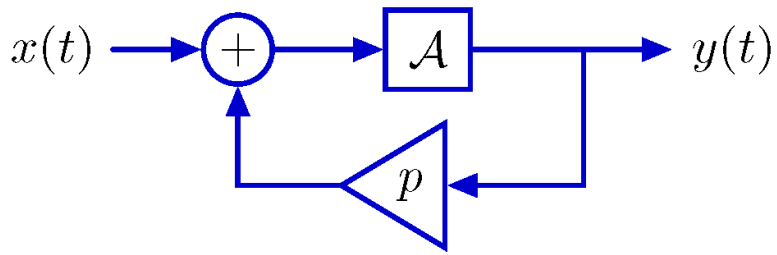

$p_0 = 1$




$p_0 = 1.2$

# Review: CT Feedback

Find the impulse response of this CT system with feedback.




Method 2: use operators.

$$Y = \mathcal{A}\,(X + pY)$$
$$\frac{Y}{X} = \frac{\mathcal{A}}{1 - p\mathcal{A}}$$

Now expand in ascending series in $\mathcal{A}$:

$$\frac{Y}{X} = \mathcal{A}(1 + p\mathcal{A} + p^2\mathcal{A}^2 + p^3\mathcal{A}^3 + \cdots)$$

If $x(t) = \delta(t)$ then

$$y(t) = \mathcal{A}(1 + p\mathcal{A} + p^2\mathcal{A}^2 + p^3\mathcal{A}^3 + \cdots)\,\delta(t)$$
$$= (1 + pt + \frac{1}{2}p^2t^2 + \frac{1}{6}p^3t^3 + \cdots)\,u(t) = e^{pt}u(t)\,.$$

# Review: Laplace Transform: Definition

Laplace transform maps a function of time $t$ to a function of $s$.

$$X(s) = \int x(t)e^{-st}dt$$

There are two important variants:

Unilateral

$$X(s) = \int_0^\infty x(t)e^{-st}dt$$

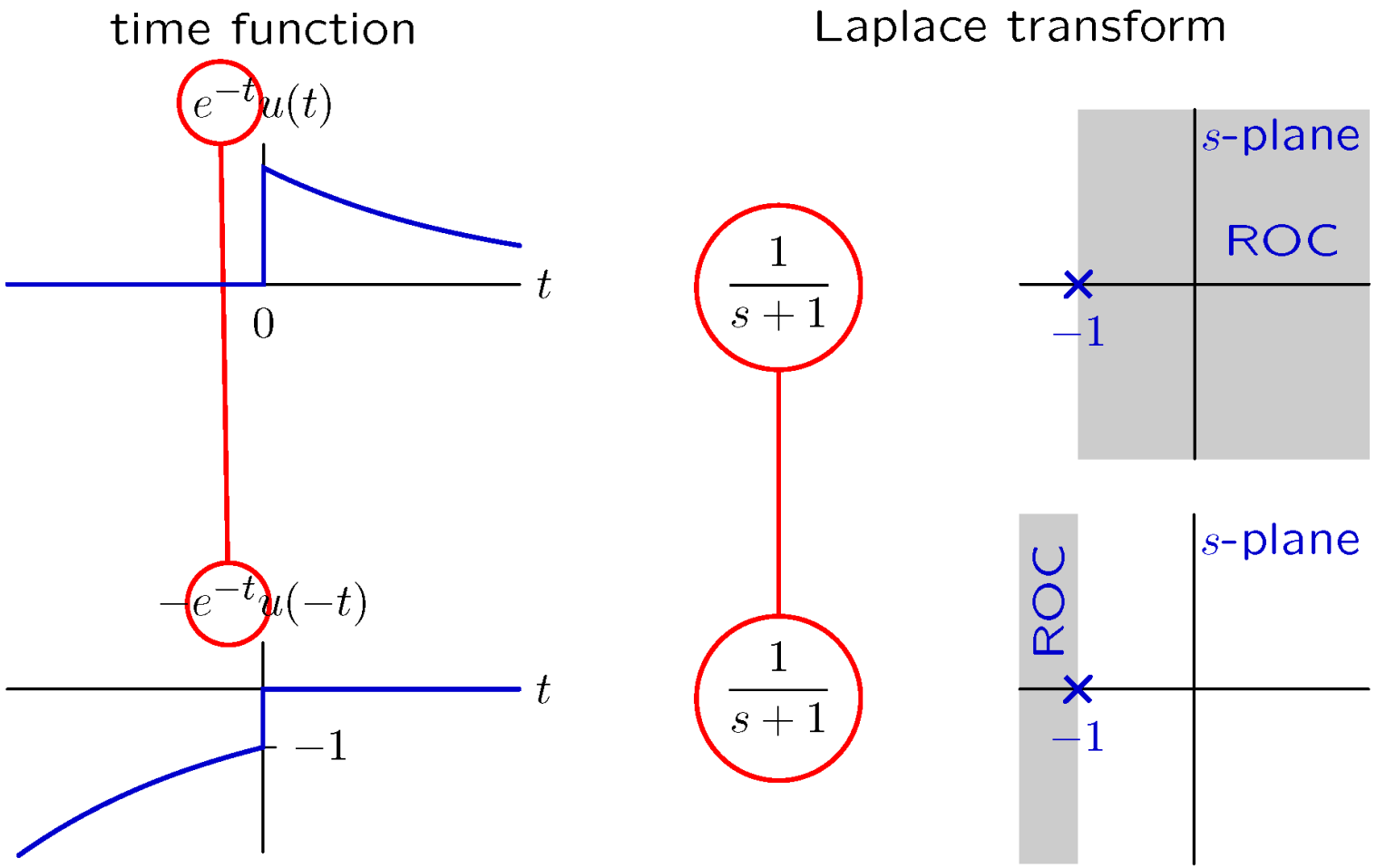
Bilateral

$$X(s) = \int_{-\infty}^\infty x(t)e^{-st}dt$$

Both share important properties — will discuss differences later.

# Review: Left- and Right-Sided ROCs

Laplace transforms of left- and right-sided exponentials have the same form (except $-$); with left- and right-sided ROCs, respectively.



# Review: Solving Differential Equations with Laplace Transforms

Summary of method.

Start with differential equation:

$$\dot{y}(t) + y(t) = \delta(t)$$

Take the Laplace transform of this equation:

$$sY(s) + Y(s) = 1$$

Solve for $Y(s)$:

$$Y(s) = \frac{1}{s+1}$$

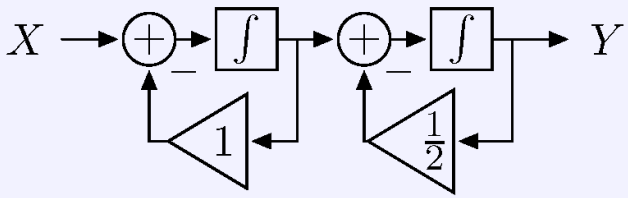
Take inverse Laplace transform (by recognizing form of transform):

$$y(t) = e^{-t}u(t)$$

# Review: Laplace Transform

Relations among representations.

**Block Diagram**




**System Functional**

$$\frac{Y}{X} = \frac{2\mathcal{A}^2}{2 + 3\mathcal{A} + \mathcal{A}^2}$$

**Impulse Response**

$$h(t) = 2(e^{-t/2} - e^{-t})\, u(t)$$

**Differential Equation**

$$2\ddot{y}(t) + 3\dot{y}(t) + y(t) = 2x(t)$$

**System Function**

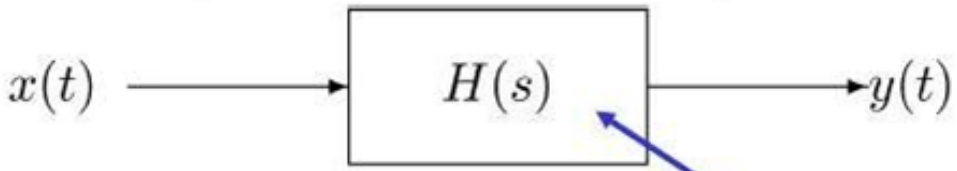
$$\frac{Y(s)}{X(s)} = \frac{2}{2s^2 + 3s + 1}$$

# Review: Laplace Transform

The use of Laplace Transforms to solve differential equations depends on several important properties.

| Property | $x(t)$ | $X(s)$ | ROC |
|---|---|---|---|
| Linearity | $ax_1(t) + bx_2(t)$ | $aX_1(s) + bX_2(s)$ | $\supset (R_1 \cap R_2)$ |
| Delay by $T$ | $x(t-T)$ | $X(s)e^{-sT}$ | $R$ |
| Multiply by $t$ | $tx(t)$ | $-\dfrac{dX(s)}{ds}$ | $R$ |
| Multiply by $e^{-\alpha t}$ | $x(t)e^{-\alpha t}$ | $X(s+\alpha)$ | shift $R$ by $-\alpha$ |
| Differentiate in $t$ | $\dfrac{dx(t)}{dt}$ | $sX(s)$ | $\supset R$ |
| Integrate in $t$ | $\displaystyle\int_{-\infty}^{t} x(\tau)\,d\tau$ | $\dfrac{X(s)}{s}$ | $\supset \Big(R \cap \big(\mathrm{Re}(s) > 0$ |
| Convolve in $t$ | $\displaystyle\int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)\,d\tau$ | $X_1(s)X_2(s)$ | $\supset (R_1 \cap R_2)$ |

# Review: CT System Function Properties




*H(s)* = "system function"

$$Y(s) = H(s)X(s)$$

1) System is stable $\Leftrightarrow \int_{-\infty}^{\infty} |h(t)|dt < \infty$ $\Leftrightarrow$ ROC of $H(s)$ includes $j\omega$ axis

2) Causality $\Rightarrow$ $h(t)$ right-sided signal $\Rightarrow$ ROC of $H(s)$ is a right-half plane

**Question:**

If the ROC of $H(s)$ is a right-half plane, is the system causal?

**Ex.** $H(s) = \dfrac{e^{sT}}{s+1}, \quad \Re e\{s\} > -1 \Rightarrow h(t)$ right-sided

$$h(t) = \mathcal{L}^{-1}\left\{\frac{e^{sT}}{s+1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}_{t\to t+T} = e^{-t}u(t)|_{t\to t+T}$$

$$= e^{-(t+T)}u(t+T) \neq 0 \quad \text{at} \quad t < 0$$ Non-causal

# Review: Properties of CT Rational System Function

a) However, if $H(s)$ is ***rational***, then

The system is causal $\Leftrightarrow$ The ROC of $H(s)$ is to the right of the rightmost pole

b) If $H(s)$ is rational and is the system function of a causal system, then

The system is stable $\Leftrightarrow$ $j\omega$-axis is in ROC
$\Leftrightarrow$ all poles are in LHP

# Initial Value Theorem

If $x(t) = 0$ for $t < 0$ and $x(t)$ contains no impulses or higher-order singularities at $t = 0$ then

$$x(0^+) = \lim_{s \to \infty} sX(s)\,.$$

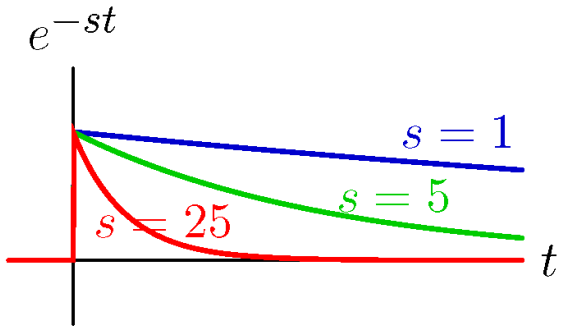
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$$x(0^+) = \lim_{s\to\infty} sX(s)\,.$$

Consider $\lim_{s\to\infty} sX(s) = \lim_{s\to\infty} s\int_{-\infty}^{\infty} x(t)e^{-st}dt = \lim_{s\to\infty}\int_0^{\infty} x(t)\,se^{-st}dt.$

As $s \to \infty$ the function $e^{-st}$ shrinks towards $0$.




Area under $e^{-st}$ is $\dfrac{1}{s}$ $\rightarrow$ area under $se^{-st}$ is $1$ $\rightarrow$ $\lim_{s\to\infty} se^{-st} = \delta(t)$ !

$$\lim_{s\to\infty} sX(s) = \lim_{s\to\infty}\int_0^{\infty} x(t)se^{-st}dt \to \int_0^{\infty} x(t)\delta(t)dt = x(0^+)$$

(the $0^+$ arises because the limit is from the right side.)

# Final Value Theorem

If $x(t) = 0$ for $t < 0$ and $x(t)$ has a finite limit as $t \to \infty$

$$x(\infty) = \lim_{s \to 0} sX(s)\,.$$

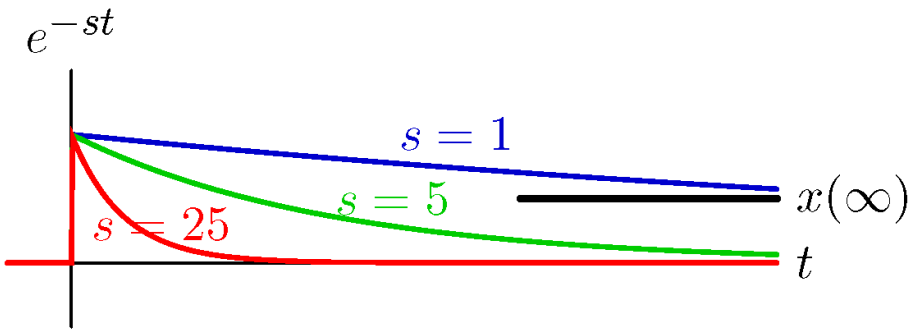
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Consider $\lim_{s \to 0} sX(s) = \lim_{s \to 0} s \int_{-\infty}^{\infty} x(t) e^{-st} dt = \lim_{s \to 0} \int_{0}^{\infty} x(t)\, se^{-st} dt.$

As $s \to 0$ the function $e^{-st}$ flattens out. But again, the area under $se^{-st}$ is always 1.




As $s \to 0$, area under $se^{-st}$ monotonically shifts to higher values of $t$ (e.g., the average value of $se^{-st}$ is $\frac{1}{s}$ which grows as $s \to 0$).

In the limit, $\lim_{s \to 0} sX(s) \to x(\infty)\,.$

# Z Transform

Z transform is discrete-time analog of Laplace transform.

# Z Transform

Z transform is discrete-time analog of Laplace transform.

Furthermore, you already know about Z transforms
(we just haven't called them Z transforms) !

Example: Fibonacci system

| | |
|---|---|
| difference equation | $y[n] = x[n] + y[n-1] + y[n-2]$ |
| operator expression | $Y = X + \mathcal{R}Y + \mathcal{R}^2 Y$ |
| system functional | $\frac{Y}{X} = \frac{1}{1 - \mathcal{R} - \mathcal{R}^2}$ |
| unit-sample response | $h[n]: \ 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \ldots$ |

# Check Yourself

Example: Fibonacci system

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What is the relation between system functional and $h[n]$?

# Check Yourself

system functional $\quad \dfrac{Y}{X} = \dfrac{1}{1 - \mathcal{R} - \mathcal{R}^2}$

unit-sample response $\quad h[n]\colon\ 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \ldots$

Expand functional in a series:

$$
\begin{array}{r|l}
 & 1 \ +\mathcal{R} \ +2\mathcal{R}^2 \ +3\mathcal{R}^3 \ +5\mathcal{R}^4 \ +8\mathcal{R}^5 \ +\cdots \\
1 - \mathcal{R} - \mathcal{R}^2 & \\
\hline
 & 1 \\
 & \underline{1 \ -\mathcal{R} \ -\mathcal{R}^2} \\
 & \quad \mathcal{R} \ +\mathcal{R}^2 \\
 & \quad \underline{\mathcal{R} \ -\mathcal{R}^2 \ -\mathcal{R}^3} \\
 & \qquad 2\mathcal{R}^2 \ +\mathcal{R}^3 \\
 & \qquad \underline{2\mathcal{R}^2 \ -2\mathcal{R}^3 \ -2\mathcal{R}^4} \\
 & \qquad\quad 3\mathcal{R}^3 \ +2\mathcal{R}^4 \\
 & \qquad\quad \underline{3\mathcal{R}^3 \ -3\mathcal{R}^4 \ -3\mathcal{R}^5} \\
 & \qquad\qquad \cdots
\end{array}
$$

$$
\begin{aligned}
\frac{Y}{X} = \frac{1}{1 - \mathcal{R} - \mathcal{R}^2} &= 1 + \mathcal{R} + 2\mathcal{R}^2 + 3\mathcal{R}^3 + 5\mathcal{R}^4 + 8\mathcal{R}^5 + 13\mathcal{R}^6 + \cdots \\
&= h[0] + h[1]\mathcal{R} + h[2]\mathcal{R}^2 + h[3]\mathcal{R}^3 + h[4]\mathcal{R}^4 + \cdots \\
&= \sum_n h[n]\mathcal{R}^n
\end{aligned}
$$

# Check Yourself

Example: Fibonacci system

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$$\frac{Y}{X} = \sum_n h[n]\mathcal{R}^n$$

What's the relation between $H(z)$ and $h[n]$?

# Check Yourself

Series expansion of system functional:

$$\frac{Y}{X} = \sum_n h[n]\mathcal{R}^n$$

Substitute $\mathcal{R} \to \frac{1}{z}$:

$$H(z) = \sum_n h[n]z^{-n}$$

# Check Yourself

Example: Fibonacci system

| | |
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$$\frac{Y}{X} = \sum_n h[n]\mathcal{R}^n$$

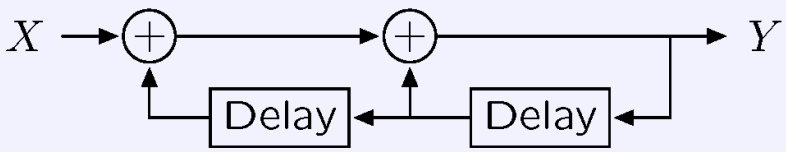
What's the relation between $H(z)$ and $h[n]$?

$$H(z) = \sum_n h[n] z^{-n}$$

# Concept Map: Discrete-Time Systems

Multiple representations of DT systems.

## Block Diagram




## System Functional

$$\frac{Y}{X} = \frac{1}{1 - \mathcal{R} - \mathcal{R}^2}$$

## Unit-Sample Response

$h[n]$: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

## Difference Equation

$$y[n] = x[n] + y[n-1] + y[n-2]$$

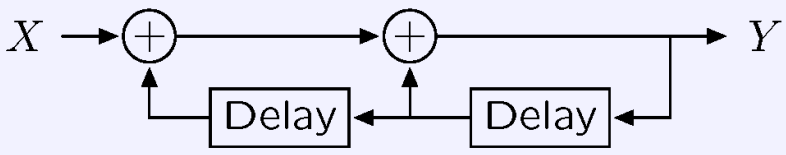
## System Function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^2}{1 - z - z^2}$$

# Concept Map: Discrete-Time Systems

Relation between Unit-Sample Response and System Functional.

### Block Diagram




### System Functional

$$\frac{Y}{X} = \frac{1}{1 - \mathcal{R} - \mathcal{R}^2}$$

series

### Unit-Sample Response

$h[n]$: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

### Difference Equation

$$y[n] = x[n] + y[n-1] + y[n-2]$$

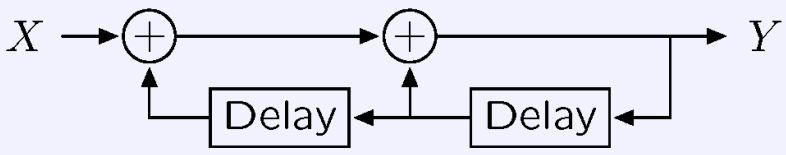
### System Function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^2}{1 - z - z^2}$$

# Concept Map: Discrete-Time Systems

Relation between System Functional and System Function.

**Block Diagram**




**System Functional**

$$\frac{Y}{X} = \frac{1}{1 - \mathcal{R} - \mathcal{R}^2}$$

$\mathcal{R} \to \frac{1}{z}$

**Unit-Sample Response**

$h[n]$: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

**Difference Equation**

$$y[n] = x[n] + y[n-1] + y[n-2]$$

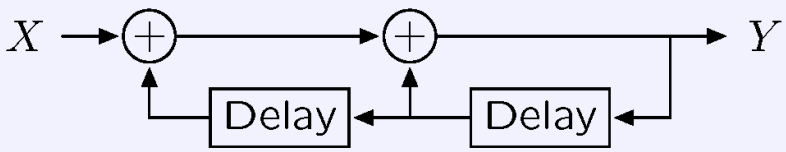
**System Function**

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^2}{1 - z - z^2}$$

# Concept Map: Discrete-Time Systems

Relation between Unit-Sample Response and System Function.

**Block Diagram**




**System Functional**

$$\frac{Y}{X} = \frac{1}{1 - \mathcal{R} - \mathcal{R}^2}$$

series

**Unit-Sample Response**

$h[n]$: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

$\mathcal{R} \to \frac{1}{z}$

Z transform

**Difference Equation**

$$y[n] = x[n] + y[n-1] + y[n-2]$$

**System Function**

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^2}{1 - z - z^2}$$

# Z Transform

Z transform is discrete-time analog of Laplace transform.

Z transform maps a function of discrete time $n$ to a function of $z$.

$$X(z) = \sum_{n} x[n] z^{-n}$$

There are two important variants:

Unilateral

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

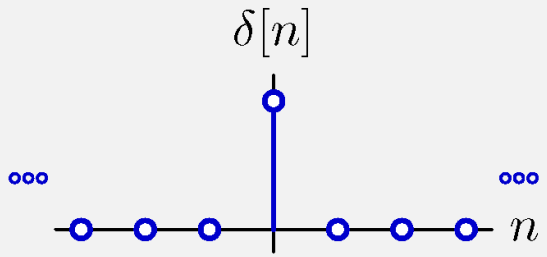
Bilateral

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Differences are analogous to those for the Laplace transform.

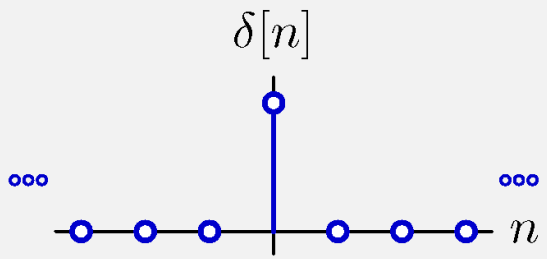
# Check Yourself

Find the Z transform of the unit-sample signal.



# Check Yourself

Find the Z transform of the unit-sample signal.



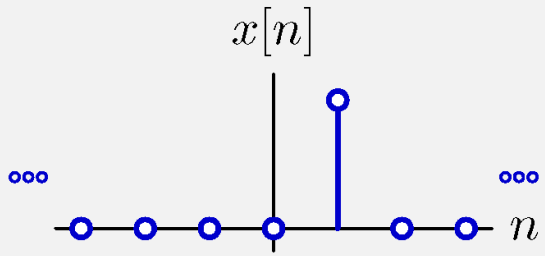

$$x[n] = \delta[n]$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = x[0]z^{0} = 1$$

$\mathcal{Z}\{\delta[n]\} = 1$, analogous to $\mathcal{L}\{\delta(t)\} = 1$.

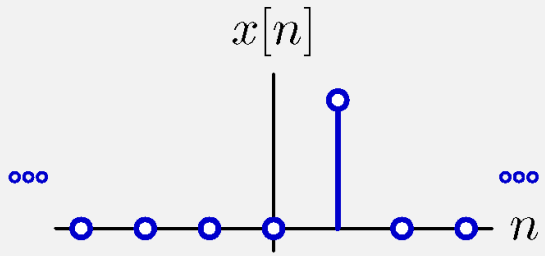
# Check Yourself

Find the Z transform of a delayed unit-sample signal.



# Check Yourself

Find the Z transform of a delayed unit-sample signal.



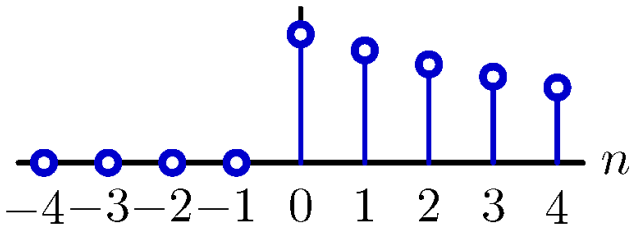

$$x[n] = \delta[n-1]$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = x[1]z^{-1} = z^{-1}$$

# Z Transforms

Example: Find the Z transform of the following signal.

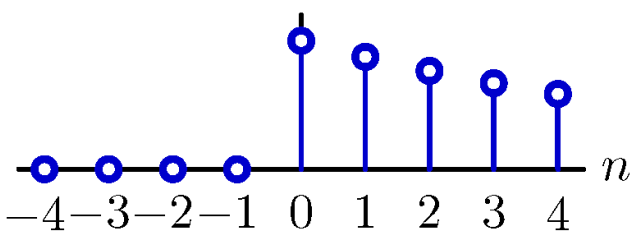
$$x[n] = \left(\frac{7}{8}\right)^n u[n]$$



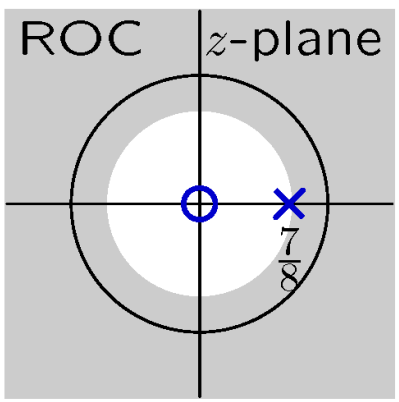

$$X(z) = \sum_{n=-\infty}^{\infty} \left(\frac{7}{8}\right)^n z^{-n} u[n] = \sum_{n=0}^{\infty} \left(\frac{7}{8}\right)^n z^{-n} = \frac{1}{1 - \frac{7}{8} z^{-1}} = \frac{z}{z - \frac{7}{8}}$$

provided $\left|\frac{7}{8} z^{-1}\right| < 1$, i.e., $|z| > \frac{7}{8}$.

$$x[n] = \left(\frac{7}{8}\right)^n u[n]$$




$$\frac{z}{z - \frac{7}{8}}$$



# Shape of ROC

Regions of converge for Z transform are delimited by circles.

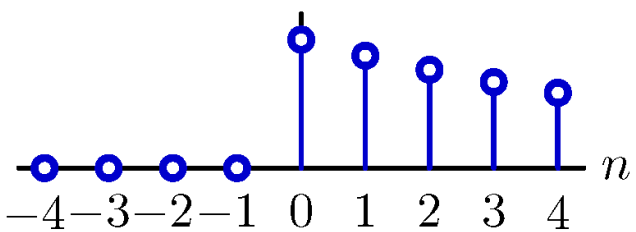
Example: $x[n] = \alpha^n u[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} \alpha^n u[n] z^{-n} = \sum_{n=0}^{\infty} \alpha^n z^{-n}$$

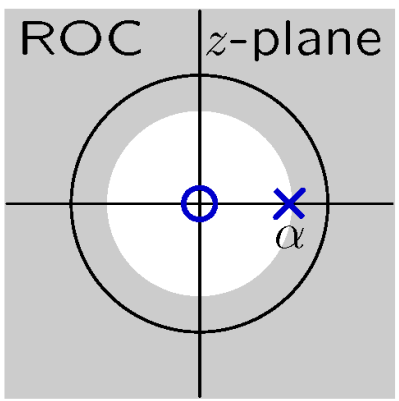
$$= \frac{1}{1 - \alpha z^{-1}}; \quad \left|\alpha z^{-1}\right| < 1$$

$$= \frac{z}{z - \alpha}; \quad |z| > |\alpha|$$

$x[n] = \alpha^n u[n]$




$$\frac{z}{z - \alpha}$$



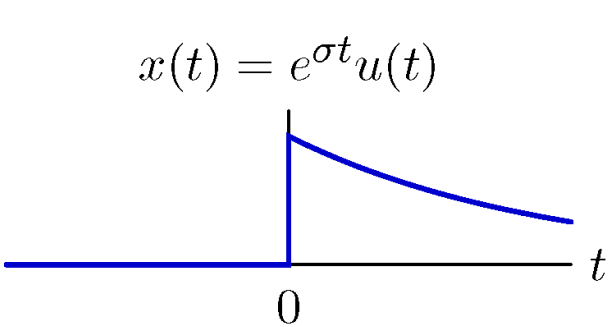
# Shape of ROC

Regions of converge for Laplace transform delimited by vertical lines.

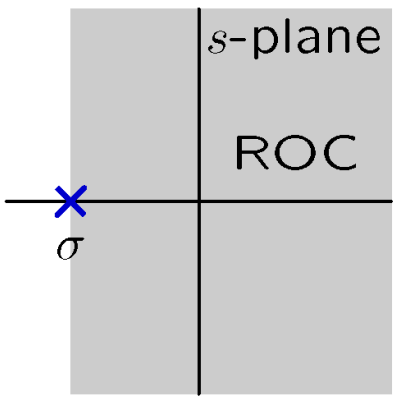
Example: $x(t) = e^{\sigma t}u(t)$

$$X(s) = \int_{-\infty}^{\infty} e^{\sigma t}u(t)e^{-st}dt = \int_{0}^{\infty} e^{\sigma t}e^{-st}dt$$

$$= \frac{1}{s-\sigma}; \quad \mathrm{Re}(s) > \mathrm{Re}(\sigma)$$




$$\frac{1}{s-\sigma}$$



# Distinguishing Features of Transforms

Most-important feature of Laplace transforms is the derivative rule:

$$x(t) \leftrightarrow X(s)$$

$$\dot{x}(t) \leftrightarrow sX(s)$$

$\rightarrow$ allows us to use Laplace transforms to solve differential equations.

Similarly, most-important feature of Z transforms is the delay rule:

$$x[n] \leftrightarrow X(z)$$

$$x[n-1] \leftrightarrow z^{-1}X(z)$$

$\rightarrow$ allows us to use Z transforms to solve difference equations.

# Distinguishing Features of Transforms

Delay property

$$x[n] \leftrightarrow X(z)$$
$$x[n-1] \leftrightarrow z^{-1}X(z)$$

Proof:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Let $y[n] = x[n-1]$ then

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n-1]z^{-n}$$

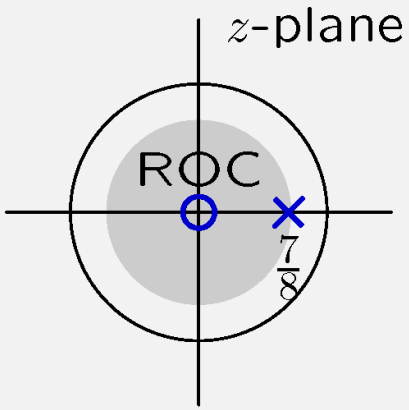
Substitute $m = n - 1$

$$Y(z) = \sum_{m=-\infty}^{\infty} x[m]z^{-m-1} = z^{-1}X(z)$$

# Check Yourself

What DT signal has the following Z transform?

$$\frac{z}{z-\frac{7}{8}}; \quad |z| < \frac{7}{8}$$



# Check Yourself

If

$$Y(z) = \frac{z}{z - \frac{7}{8}}\,; \quad |z| < \frac{7}{8}$$

then $y[n]$ corresponds to the unit-sample response of

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z}{z - \frac{7}{8}}\,.$$

The difference equation for this system is

$$y[n+1] - \frac{7}{8}y[n] = x[n+1]\,.$$

Convergence **inside** $|z| = \frac{7}{8}$ corresponds to a left-sided (non-causal) response. Solve by iterating backwards in time:

$$y[n] = \frac{8}{7}\,\left(y[n+1] - x[n+1]\right)$$

# Check Yourself

Solve by iterating backwards in time:

$$y[n] = \frac{8}{7}\left(y[n+1] - x[n+1]\right)$$

Start "at rest":

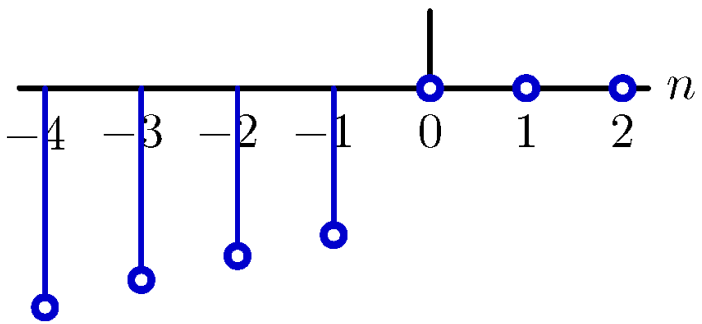
| $n$ | $x[n]$ | $y[n]$ |
|---|---|---|
| $> 0$ | $0$ | $0$ |
| $0$ | $1$ | $0$ |
| $-1$ | $0$ | $-\left(\frac{8}{7}\right)$ |
| $-2$ | $0$ | $-\left(\frac{8}{7}\right)^2$ |
| $-3$ | $0$ | $-\left(\frac{8}{7}\right)^3$ |
| $\cdots$ | | $\cdots$ |
| $n$ | | $-\left(\frac{8}{7}\right)^{-n}$ |

$$y[n] = -\left(\frac{8}{7}\right)^{-n} \;; \quad n < 0 = -\left(\frac{7}{8}\right)^{n} u[-1-n]$$

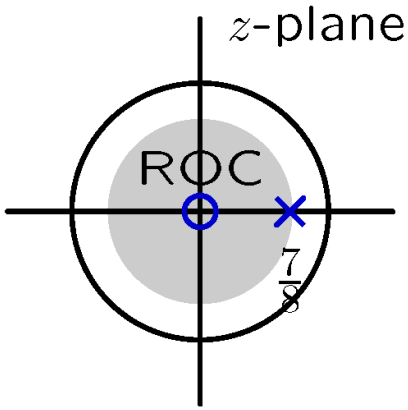
# Check Yourself

Plot

$$y[n] = -\left(\frac{7}{8}\right)^n u[-1-n]$$



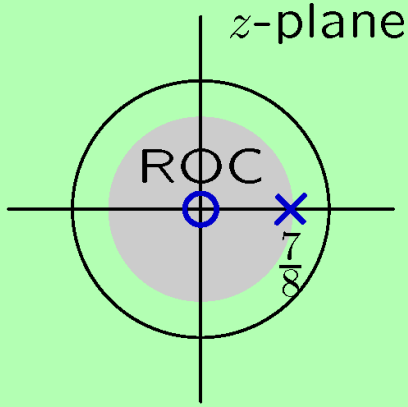

$$\frac{z}{z-\frac{7}{8}}$$



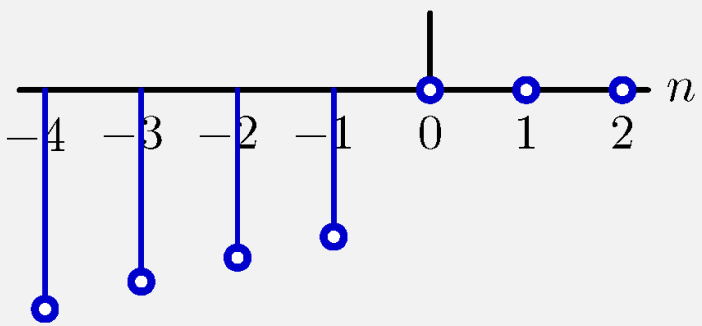
# Check Yourself

What DT signal has the following Z transform?

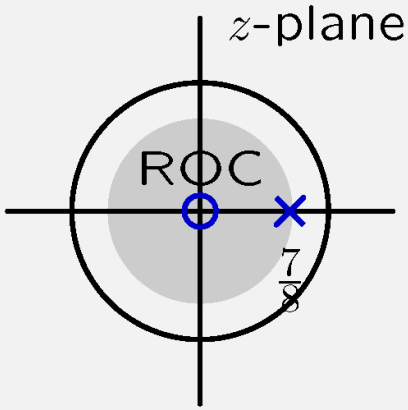
$$\frac{z}{z-\frac{7}{8}}; \quad |z| < \frac{7}{8}$$




$$y[n] = -\left(\frac{7}{8}\right)^n u[-1-n]$$




$$\frac{z}{z-\frac{7}{8}}$$



# Check Yourself

Find the inverse transform of

$$X(z) = \frac{-3z}{2z^2 - 5z + 2}$$

given that the ROC includes the unit circle.

# Check Yourself

Find the inverse transform of

$$X(z) = \frac{-3z}{2z^2 - 5z + 2}$$

given that the ROC includes the unit circle.

Expand with partial fractions:

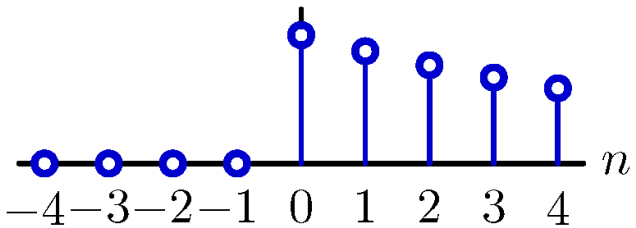
$$X(z) = \frac{-3z}{2z^2 - 5z + 2} = \frac{1}{2z - 1} - \frac{2}{z - 2}$$

Not at standard form!

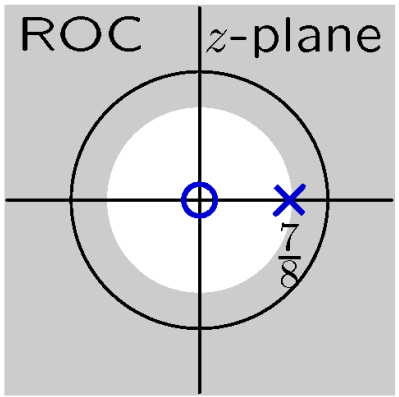
# Check Yourself

Standard forms:

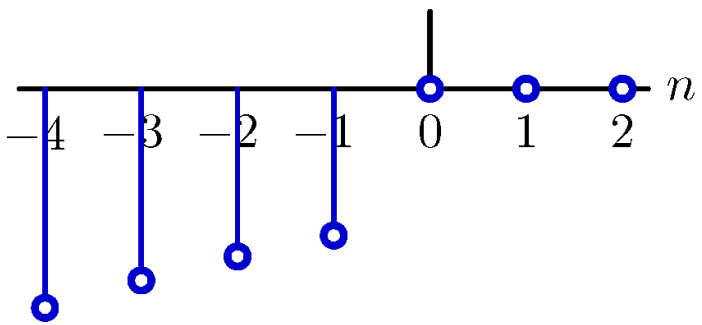
$$x[n] = \left(\frac{7}{8}\right)^n u[n]$$



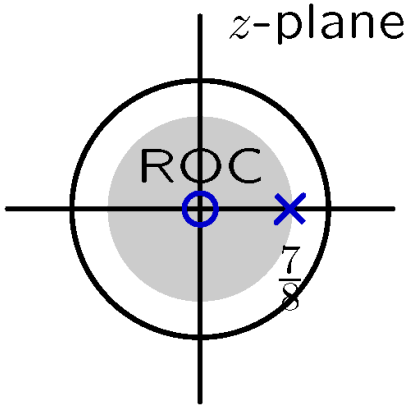

$$\frac{z}{z - \frac{7}{8}}$$




$$y[n] = -\left(\frac{7}{8}\right)^n u[-1-n]$$




$$\frac{z}{z - \frac{7}{8}}$$



# Check Yourself

Find the inverse transform of

$$X(z) = \frac{-3z}{2z^2 - 5z + 2}$$

given that the ROC includes the unit circle.

Expand with partial fractions:

$$X(z) = \frac{-3z}{2z^2 - 5z + 2} = \frac{1}{2z - 1} - \frac{2}{z - 2}$$

Not at standard form!

Expand it differently: as a standard form:

$$X(z) = \frac{-3z}{2z^2 - 5z + 2} = \frac{2z}{2z - 1} - \frac{z}{z - 2} = \frac{z}{z - \frac{1}{2}} - \frac{z}{z - 2}$$

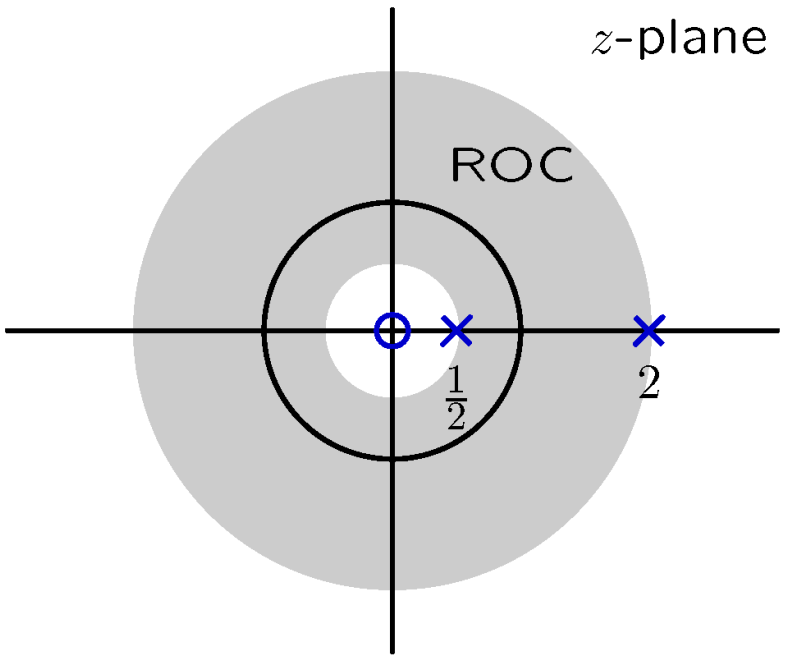
Standard form: a pole at $\frac{1}{2}$ and a pole at $2$.

# Check Yourself

Ratio of polynomials in $z$:

$$X(z) = \frac{-3z}{2z^2 - 5z + 2} = \frac{z}{z - \frac{1}{2}} - \frac{z}{z - 2}$$

– a pole at $\frac{1}{2}$ and a pole at $2$.



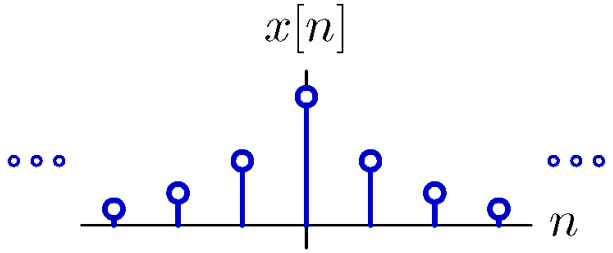

Region of convergence is "outside" pole at $\frac{1}{2}$ but "inside" pole at $2$.

$$x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-1 - n]$$

# Check Yourself

Plot.

$$x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-1-n]$$



# Check Yourself

Alternatively, stick with non-standard form:

$$X(z) = \frac{-3z}{2z^2 - 5z + 2} = \frac{1}{2z - 1} - \frac{2}{z - 2}$$

Make it look more standard:

$$X(z) = \frac{1}{2} z^{-1} \frac{z}{z - \frac{1}{2}} - 2z^{-1} \frac{z}{z - 2}$$

# Check Yourself

Alternatively, stick with non-standard form:

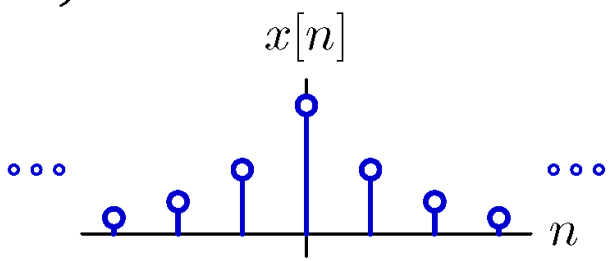
$$X(z) = \frac{-3z}{2z^2 - 5z + 2} = \frac{1}{2z - 1} - \frac{2}{z - 2}$$

Make it look more standard:

$$X(z) = \frac{1}{2} z^{-1} \frac{z}{z - \frac{1}{2}} - 2z^{-1} \frac{z}{z - 2}$$

Now

$$\begin{aligned} x[n] &= \frac{1}{2} \mathcal{R} \left\{ \left( \frac{1}{2} \right)^n u[n] \right\} + 2\mathcal{R} \left\{ +2^n u[-1 - n] \right\} \\ &= \frac{1}{2} \left\{ \left( \frac{1}{2} \right)^{n-1} u[n - 1] \right\} + 2 \left\{ +2^{n-1} u[-n] \right\} \\ &= \left\{ \left( \frac{1}{2} \right)^n u[n - 1] \right\} + \left\{ +2^n u[-n] \right\} \end{aligned}$$



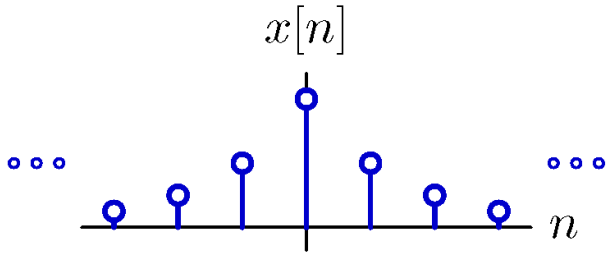
# Check Yourself

Alternative 3: expand as polynomials in $z^{-1}$:

$$X(z) = \frac{-3z}{2z^2 - 5z + 2} = \frac{-3z^{-1}}{2 - 5z^{-1} + 2z^{-2}}$$
$$= \frac{2}{2 - z^{-1}} - \frac{1}{1 - 2z^{-1}} = \frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{1}{1 - 2z^{-1}}$$

Now

$$x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-1 - n]$$

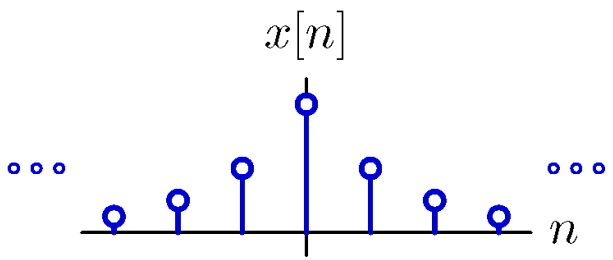


# Check Yourself

Find the inverse transform of

$$X(z) = \frac{-3z}{2z^2 - 5z + 2}$$

given that the ROC includes the unit circle.



# Solving Difference Equations with Z Transforms

Start with difference equation:

$$y[n] - \frac{1}{2}y[n-1] = \delta[n]$$

Take the Z transform of this equation:

$$Y(z) - \frac{1}{2}z^{-1}Y(z) = 1$$

Solve for $Y(z)$:

$$Y(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

Take the inverse Z transform (by recognizing the form of the transform):

$$y[n] = \left(\frac{1}{2}\right)^n u[n]$$

# Inverse Z transform

The inverse Z transform is defined by an integral that is not particularly easy to solve.

Formally,

$$x[n] = \frac{1}{2\pi j} \int_C X(z) z^{n-1} ds$$

were $C$ represents a closed contour that circles the origin by running in a counterclockwise direction through the region of convergence. This integral is not generally easy to compute.

This equation can be useful to prove theorems.

There are better ways (e.g., partial fractions) to compute inverse transforms for the kinds of systems that we frequently encounter.

# Properties of Z Transforms

The use of Z Transforms to solve differential equations depends on several important properties.

| Property | $x[n]$ | $X(z)$ | ROC |
|---|---|---|---|
| Linearity | $ax_1[n] + bx_2[n]$ | $aX_1(z) + bX_2(z)$ | $\supset (R_1 \cap R_2)$ |
| Delay | $x[n-1]$ | $z^{-1}X(z)$ | $R$ |
| Multiply by $n$ | $nx[n]$ | $-z\frac{dX(z)}{dz}$ | $R$ |
| Convolve in $n$ | $\sum_{m=-\infty}^{\infty} x_1[m]x_2[n-m]$ | $X_1(z)X_2(z)$ | $\supset (R_1 \cap R_2)$ |

# Properties of Z Transforms

**TABLE 10.1** PROPERTIES OF THE $z$-TRANSFORM

| Section | Property | Signal | $z$-Transform | ROC |
|---|---|---|---|---|
| | | $x[n]$ | $X(z)$ | $R$ |
| | | $x_1[n]$ | $X_1(z)$ | $R_1$ |
| | | $x_2[n]$ | $X_2(z)$ | $R_2$ |
| 10.5.1 | Linearity | $ax_1[n] + bx_2[n]$ | $aX_1(z) + bX_2(z)$ | At least the intersection of $R_1$ and $R_2$ |
| 10.5.2 | Time shifting | $x[n - n_0]$ | $z^{-n_0}X(z)$ | $R$, except for the possible addition or deletion of the origin |
| 10.5.3 | Scaling in the $z$-domain | $e^{j\omega_0 n}x[n]$ | $X(e^{-j\omega_0}z)$ | $R$ |
| | | $z_0^n x[n]$ | $X\left(\frac{z}{z_0}\right)$ | $z_0 R$ |
| | | $a^n x[n]$ | $X(a^{-1}z)$ | Scaled version of $R$ (i.e., $\lvert a\rvert R$ = the set of points $\{\lvert a\rvert z\}$ for $z$ in $R$) |
| 10.5.4 | Time reversal | $x[-n]$ | $X(z^{-1})$ | Inverted $R$ (i.e., $R^{-1}$ = the set of points $z^{-1}$, where $z$ is in $R$) |
| 10.5.5 | Time expansion | $x_{(k)}[n] = \begin{cases} x[r], & n = rk \\ 0, & n \neq rk \end{cases}$ for some integer $r$ | $X(z^k)$ | $R^{1/k}$ (i.e., the set of points $z^{1/k}$, where $z$ is in $R$) |
| 10.5.6 | Conjugation | $x^*[n]$ | $X^*(z^*)$ | $R$ |
| 10.5.7 | Convolution | $x_1[n] * x_2[n]$ | $X_1(z)X_2(z)$ | At least the intersection of $R_1$ and $R_2$ |
| 10.5.7 | First difference | $x[n] - x[n-1]$ | $(1 - z^{-1})X(z)$ | At least the intersection of $R$ and $\lvert z\rvert > 0$ |
| 10.5.7 | Accumulation | $\sum_{k=-\infty}^{n} x[k]$ | $\frac{1}{1 - z^{-1}}X(z)$ | At least the intersection of $R$ and $\lvert z\rvert > 1$ |
| 10.5.8 | Differentiation in the $z$-domain | $nx[n]$ | $-z\frac{dX(z)}{dz}$ | $R$ |
| 10.5.9 | | Initial Value Theorem<br>If $x[n] = 0$ for $n < 0$, then<br>$x[0] = \lim_{z\to\infty} X(z)$ | | |

# Check Yourself

Find the inverse transform of $Y(z) = \left(\frac{z}{z-1}\right)^2 ; \quad |z| > 1.$

# Check Yourself

Find the inverse transform of $Y(z) = \left(\frac{z}{z-1}\right)^2$ ; $|z| > 1$.

$y[n]$ corresponds to unit-sample response of the **right-sided** system

$$\frac{Y}{X} = \left(\frac{z}{z-1}\right)^2 = \left(\frac{1}{1-z^{-1}}\right)^2 = \left(\frac{1}{1-\mathcal{R}}\right)^2$$

$$= \left(1 + \mathcal{R} + \mathcal{R}^2 + \mathcal{R}^3 + \cdots\right) \times \left(1 + \mathcal{R} + \mathcal{R}^2 + \mathcal{R}^3 + \cdots\right)$$

| | $1$ | $\mathcal{R}$ | $\mathcal{R}^2$ | $\mathcal{R}^3$ | $\cdots$ |
|---|---|---|---|---|---|
| $1$ | $1$ | $\mathcal{R}$ | $\mathcal{R}^2$ | $\mathcal{R}^3$ | $\cdots$ |
| $\mathcal{R}$ | $\mathcal{R}$ | $\mathcal{R}^2$ | $\mathcal{R}^3$ | $\mathcal{R}^4$ | $\cdots$ |
| $\mathcal{R}^2$ | $\mathcal{R}^2$ | $\mathcal{R}^3$ | $\mathcal{R}^4$ | $\mathcal{R}^5$ | $\cdots$ |
| $\mathcal{R}^3$ | $\mathcal{R}^3$ | $\mathcal{R}^4$ | $\mathcal{R}^5$ | $\mathcal{R}^6$ | $\cdots$ |
| $\cdots$ | $\cdots$ | $\cdots$ | $\cdots$ | $\cdots$ | $\cdots$ |

$$\frac{Y}{X} = 1 + 2\mathcal{R} + 3\mathcal{R}^2 + 4\mathcal{R}^3 + \cdots = \sum_{n=0}^{\infty}(n+1)\mathcal{R}^n$$

$$y[n] = h[n] = (n+1)u[n]$$

# Check Yourself

Table lookup method.

$$Y(z) = \left(\frac{z}{z-1}\right)^2 \quad \leftrightarrow \quad y[n] = ?$$

$$\frac{z}{z-1} \quad \leftrightarrow \quad u[n]$$

# Properties of Z Transforms

The use of Z Transforms to solve differential equations depends on several important properties.

| Property | $x[n]$ | $X(z)$ | ROC |
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| Convolve in $n$ | $\sum_{m=-\infty}^{\infty} x_1[m]x_2[n-m]$ | $X_1(z)X_2(z)$ | $\supset (R_1 \cap R_2)$ |

# Check Yourself

Table lookup method.

$$Y(z) = \left(\frac{z}{z-1}\right)^2 \quad \leftrightarrow \quad y[n] = ?$$

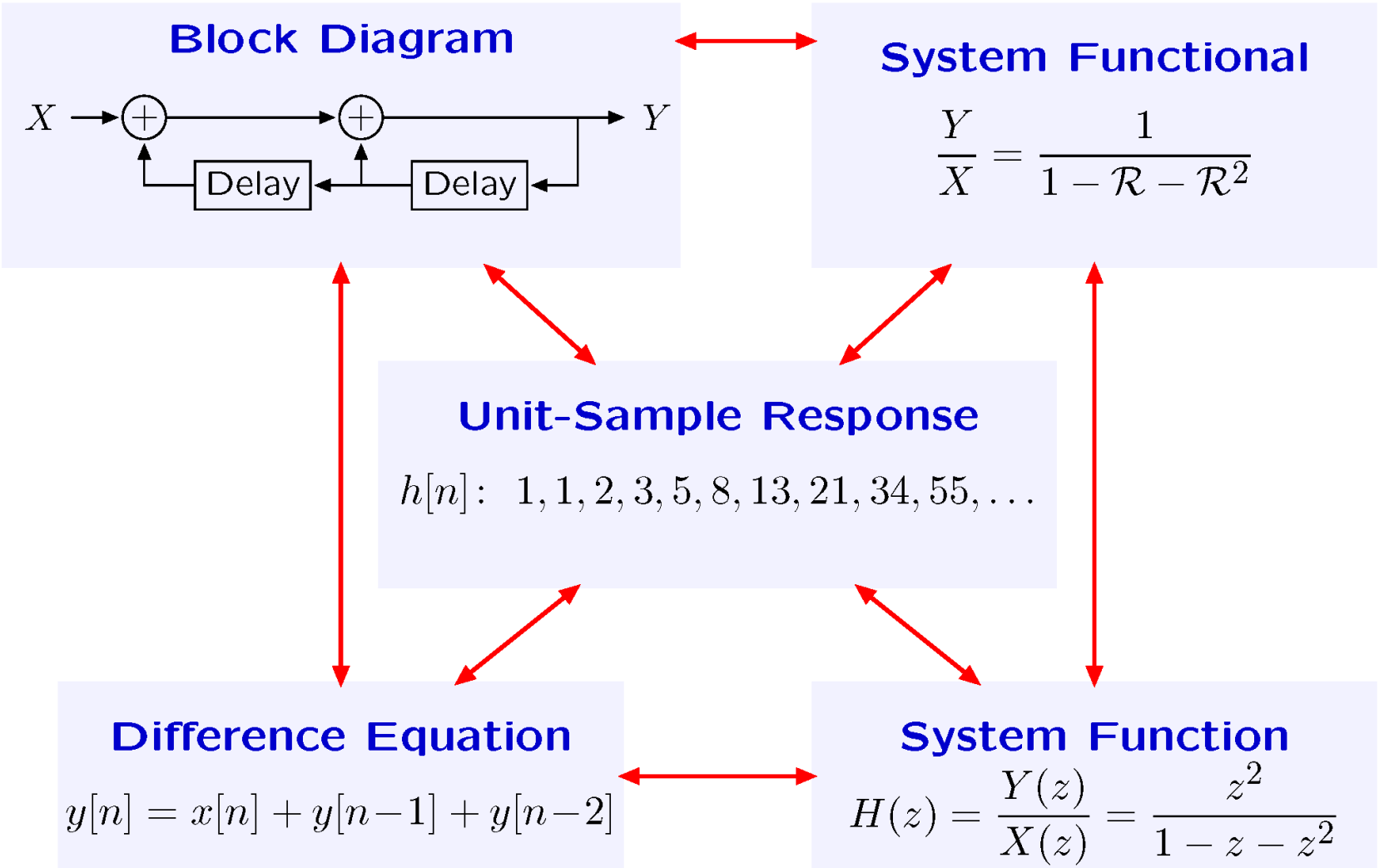
$$\frac{z}{z-1} \quad \leftrightarrow \quad u[n]$$

$$-z\frac{d}{dz}\left(\frac{z}{z-1}\right) = z\left(\frac{1}{z-1}\right)^2 \quad \leftrightarrow \quad nu[n]$$

$$z \times \left(-z\frac{d}{dz}\left(\frac{z}{z-1}\right)\right) = \left(\frac{z}{z-1}\right)^2 \quad \leftrightarrow \quad (n+1)u[n+1] = (n+1)u[n]$$

# Concept Map: Discrete-Time Systems

Relations among representations.



# The z-Transform - DT Laplace Transform

Recall:

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt \overset{t=n\Delta t}{=} \lim_{\Delta t \to 0} \sum_{n=-\infty}^{\infty} \underbrace{x(n\Delta t)\cdot \Delta t}_{x[n]} \underbrace{(e^{s\Delta t})^{-n}}_{z}$$

$\Downarrow$ DT, $\Delta t$ is now finite

$$\boxed{X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}}$$

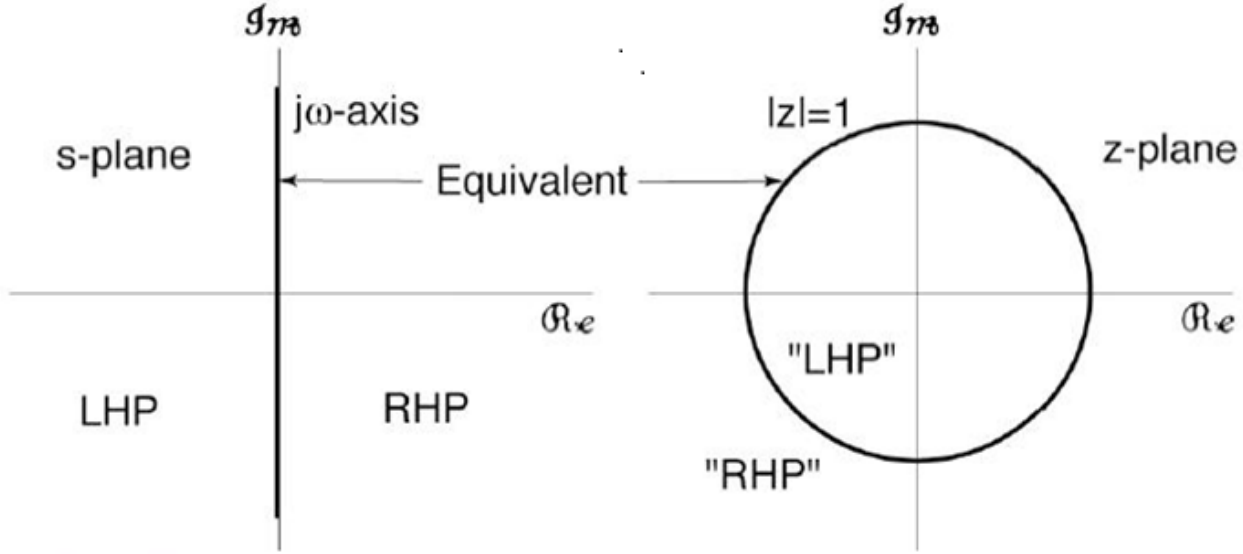
where $z = e^{s\Delta t}$

$\Downarrow$

# The Relations Between z-Transform and Laplace Transform

$$e^{s\Delta t} = z$$

$j\omega$-axis in $s$-plane ($s = j\omega$) $\Leftrightarrow$ $|z| = |e^{j\omega\Delta t}| = 1$ — a unit circle in $z$-plane




- A vertical line in $s$-plane, $Re(s)$ = constant $\Leftrightarrow$ $|e^{s\Delta t}|$ = constant, a circle in $z$-plane.
- LHP in $s$-plane, $Re(s) < 0 \Rightarrow |z| = |e^{s\Delta t}| < 1$, inside the $|z| = 1$ circle, special case, $Re(s) = -\infty \Leftrightarrow |z| = 0$.
- RHP in $s$-plane, $Re(s) > 0 \Rightarrow |z| = |e^{s\Delta t}| > 1$, outside the $|z| = 1$ circle, special case, $Re(s) = +\infty \Leftrightarrow |z| = \infty$.

# Relation Between Causal DT Time Functions and Pole-zero Diagrams

## Laplace Transform

| Pole characteristics | Time function |
|---|---|
| On real axis | Exponential |
| On imaginary axis | Sinusoid |
| In complex $s$ plane | Exponentially modulated sinusoid |
| Negative real part | Bounded |
| Positive real parts | Unbounded |
| Far from origin of $s$ plane | Rapid time course |

## Z Transform

| Pole characteristics | Time function |
|---|---|
| On positive real axis | Geometric |
| On negative real axis | Alternating sign geometric |
| On unit circle | Sinusoidal |
| In complex $z$ plane | Geometrically modulated sinusoid |
| Inside unit circle | Bounded |
| Outside unit circle | Unbounded |
| Far from $z = 1$ | Rapid time course |

# DT LTI Systems Described by LCCDEs

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

Use the time-shift property

$$\sum_{k=0}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z)$$

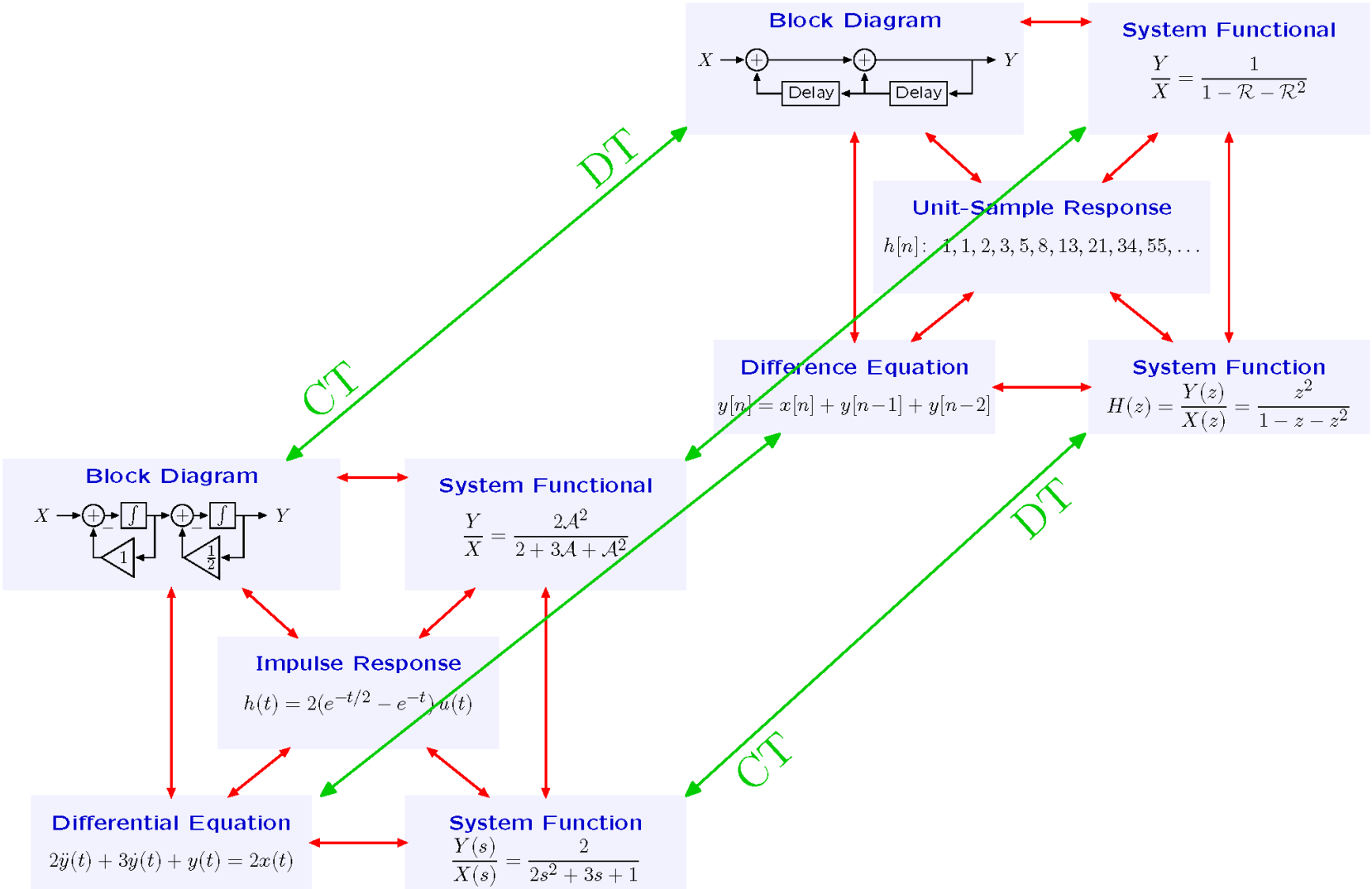
$$\Downarrow$$

$$Y(z) = H(z)X(z)$$

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} \quad \text{— Rational}$$

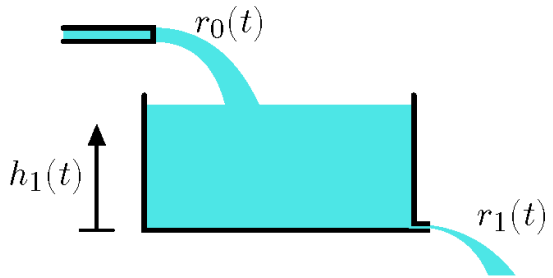
# Concept Map

Relations between CT and DT representations.

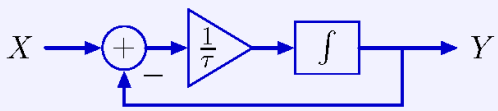


# First-Order CT System

Example: leaky tank.




**Block Diagram**




**System Functional**

$$\frac{Y}{X} = \frac{\mathcal{A}}{\mathcal{A}+\tau}$$

**Impulse Response**

$$h(t) = \tfrac{1}{\tau} e^{-t/\tau} u(t)$$

**Differential Equation**

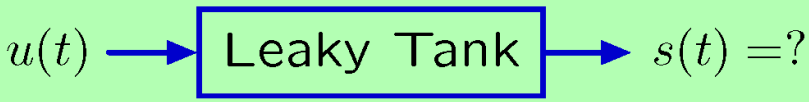
$$\tau \dot{r}_1(t) = r_0(t) - r_1(t)$$

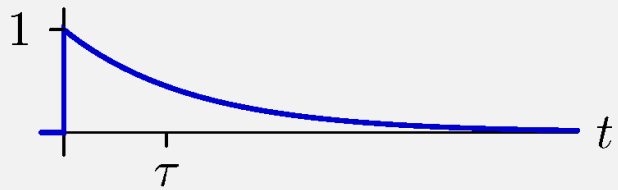
**System Function**

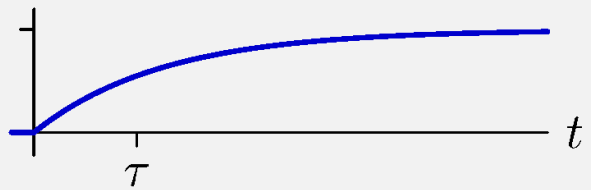
$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{1+\tau s}$$

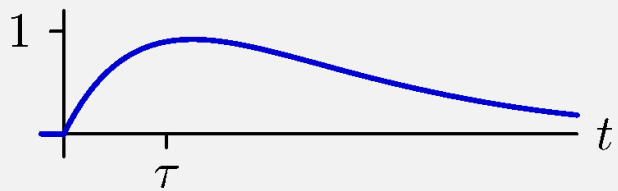
# Check Yourself

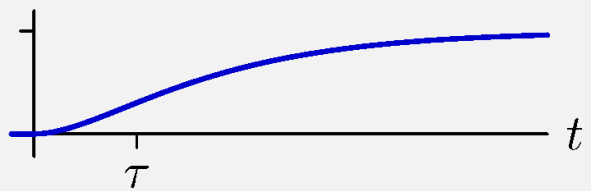
What is the "step response" of the leaky tank system?




1.



2.


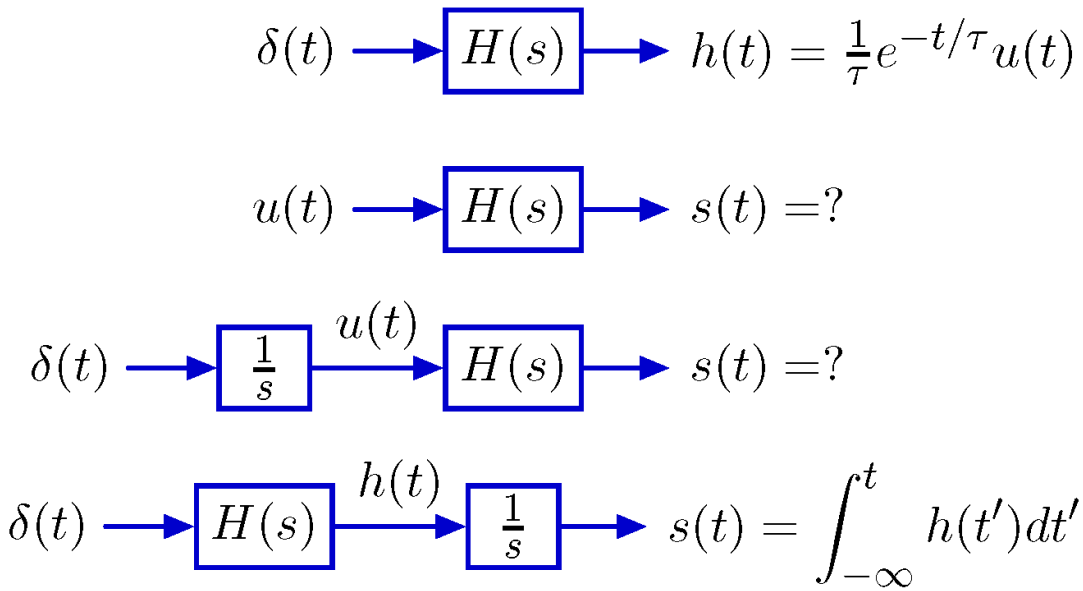

3.



4.



5. none of the above

# Check Yourself

What is the "step response" of the leaky tank system?

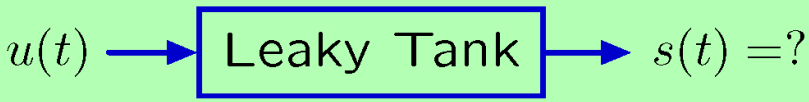


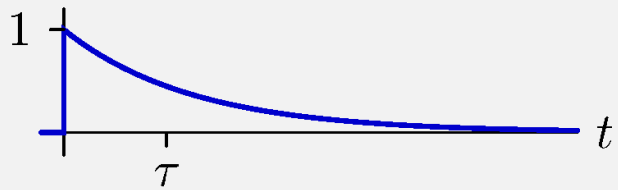

$$s(t) = \int_{-\infty}^{t} \frac{1}{\tau}e^{-t'/\tau}u(t')dt' = \int_{0}^{t} \frac{1}{\tau}e^{-t'/\tau}dt' = (1 - e^{-t/\tau})\,u(t)$$

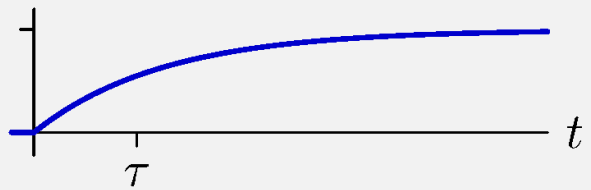
Reasoning with systems.

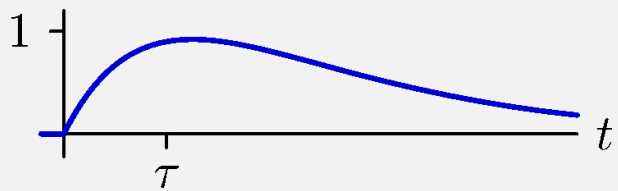
# Check Yourself

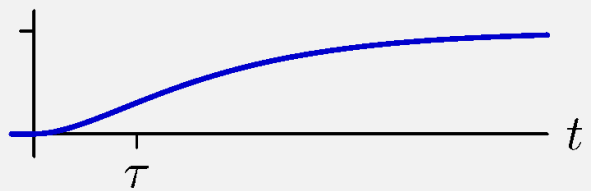
What is the "step response" of the leaky tank system? 2




1.



2.



3.



4.



5. none of the above

# Forward Euler Approximation

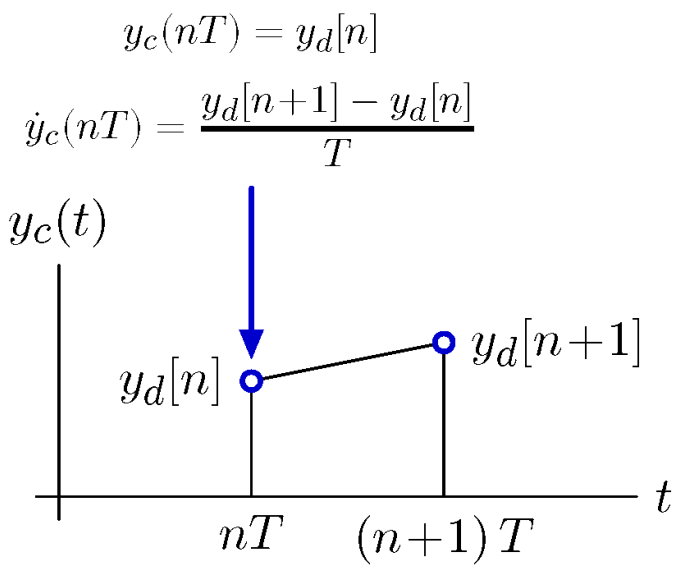
Approximate leaky-tank response using forward Euler approach.

Substitute

$$x_d[n] = x_c(nT)$$

$$y_d[n] = y_c(nT)$$

$$\dot{y}_c(nT) \approx \frac{y_c\big((n+1)T\big) - y_c\big(nT\big)}{T} = \frac{y_d[n+1] - y_d[n]}{T}$$



# Forward Euler Approximation

Approximate leaky-tank response using forward Euler approach.

Substitute

$$x_d[n] = x_c(nT)$$
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$$\dot{y}_c(nT) \approx \frac{y_c\big((n+1)T\big) - y_c\big(nT\big)}{T} = \frac{y_d[n+1] - y_d[n]}{T}$$

into the differential equation

$$\tau \dot{y}_c(t) = x_c(t) - y_c(t)$$

to obtain

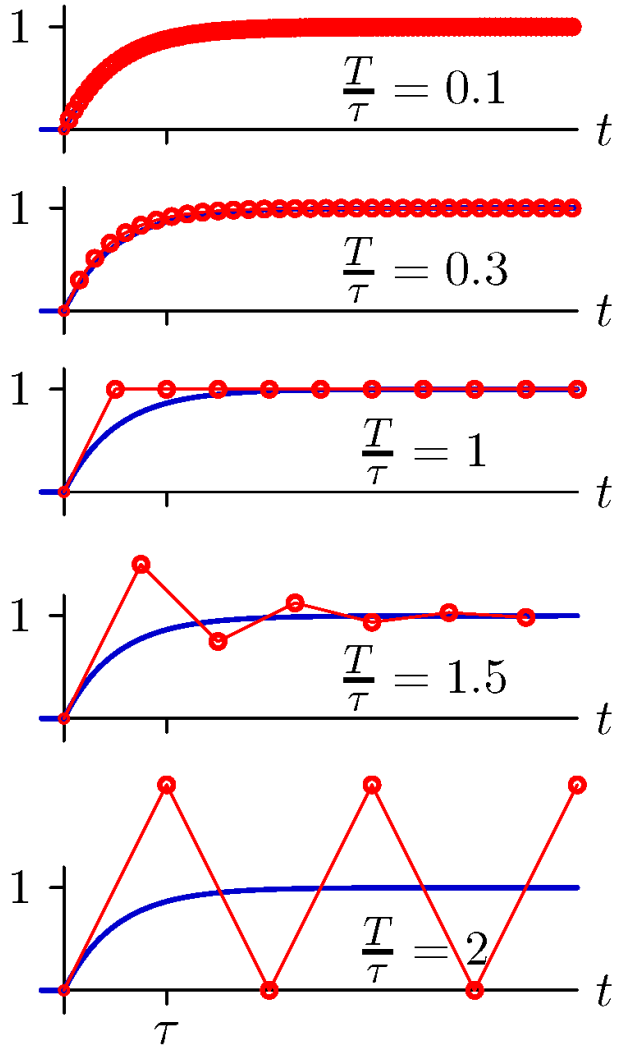
$$\frac{\tau}{T}\Big(y_d[n+1] - y_d[n]\Big) = x_d[n] - y_d[n]\,.$$

Solve:

$$y_d[n+1] - \left(1 - \frac{T}{\tau}\right) y_d[n] = \frac{T}{\tau} x_d[n]$$

# Forward Euler Approximation

Plot.




Why is this approximation badly behaved?

# Check Yourself

DT approximation:

$$y_d[n+1] - \left(1 - \frac{T}{\tau}\right) y_d[n] = \frac{T}{\tau} x_d[n]$$

Find the DT pole.

1. $z = \frac{T}{\tau}$
2. $z = 1 - \frac{T}{\tau}$
3. $z = \frac{\tau}{T}$
4. $z = -\frac{\tau}{T}$
5. $z = \frac{1}{1 + \frac{T}{\tau}}$

# Check Yourself

DT approximation:

$$y_d[n+1] - \left(1 - \frac{T}{\tau}\right) y_d[n] = \frac{T}{\tau} x_d[n]$$

Take the Z transform:

$$zY_d(z) - \left(1 - \frac{T}{\tau}\right) Y_d(z) = \frac{T}{\tau} X_d(z)$$

Solve for the system function:

$$H(z) = \frac{Y_d(z)}{X_d(z)} = \frac{\frac{T}{\tau}}{z - \left(1 - \frac{T}{\tau}\right)}$$

Pole at $z = 1 - \frac{T}{\tau}$.

# Check Yourself

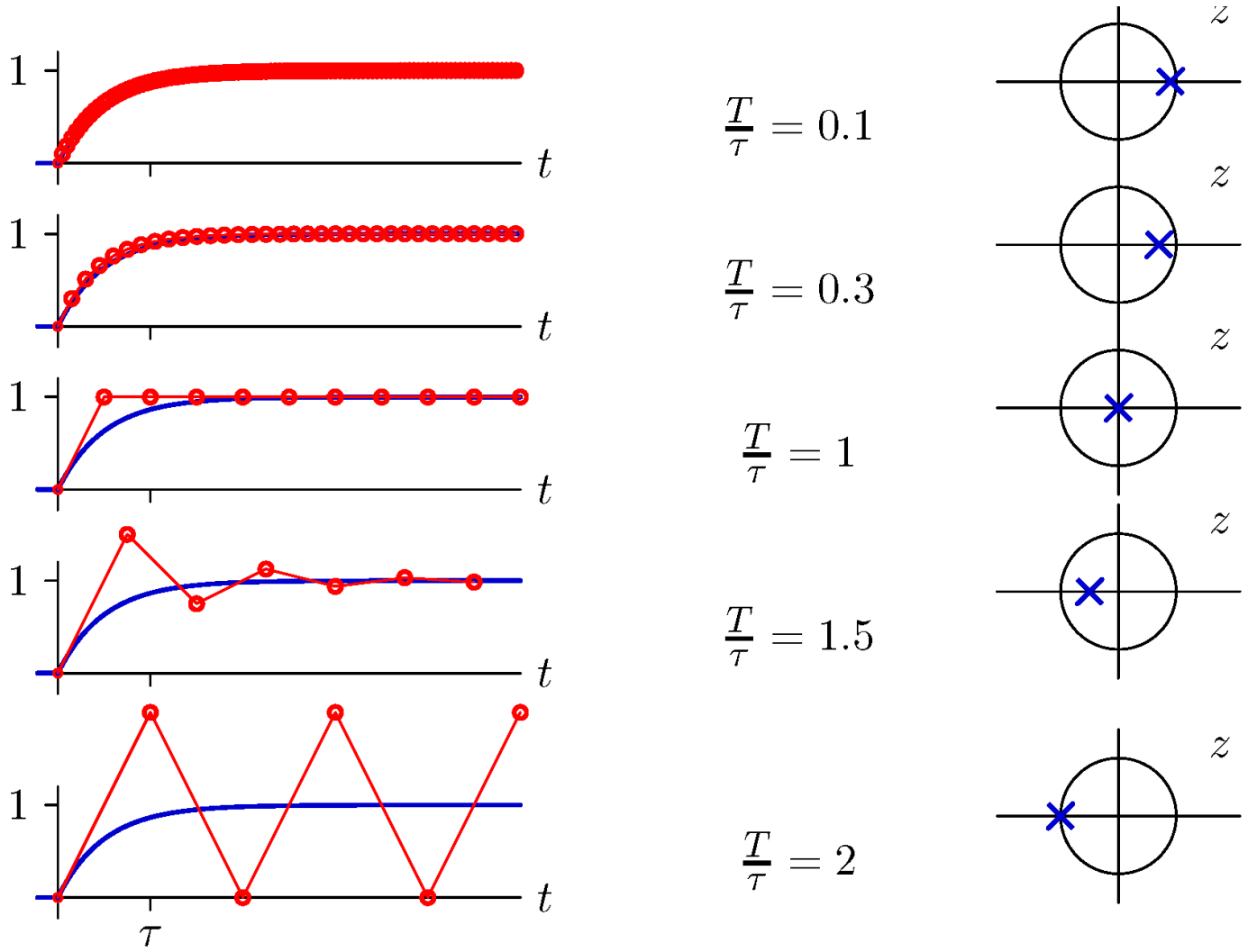
DT approximation:

$$y_d[n+1] - \left(1 - \frac{T}{\tau}\right) y_d[n] = \frac{T}{\tau} x_d[n]$$

Find the DT pole. 2

1. $z = \frac{T}{\tau}$
2. $z = 1 - \frac{T}{\tau}$
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5. $z = \frac{1}{1 + \frac{T}{\tau}}$

# Dependence of DT pole on Stepsize



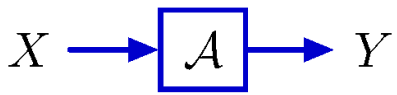

The CT pole was fixed ($s = -\frac{1}{\tau}$). Why is the DT pole changing?

# Dependence of DT pole on Stepsize

Change in DT pole: problem specific or property of forward Euler?

Approach: make a systems model of forward Euler method.

CT block diagrams: adders, gains, and integrators:

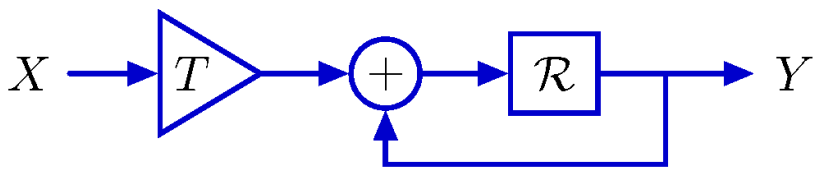



$$\dot{y}(t) = x(t)$$

Forward Euler approximation:

$$\frac{y[n+1] - y[n]}{T} = x[n]$$

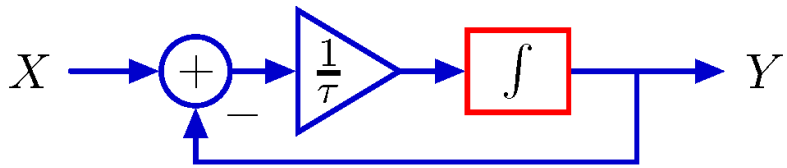
Equivalent system:



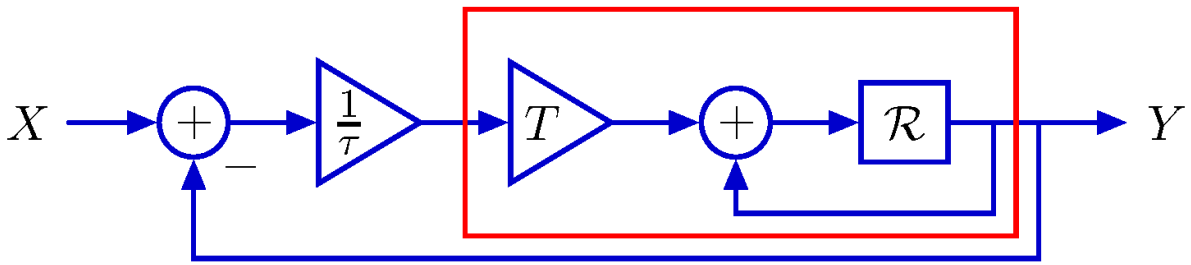

Forward Euler: substitute equivalent system for all integrators.

# Example: leaky tank system

Started with leaky tank system:




Replace integrator with forward Euler rule:




Write system functional:

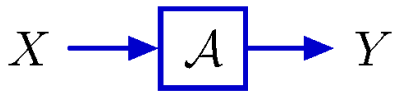
$$\frac{Y}{X} = \frac{\frac{T}{\tau}\frac{\mathcal{R}}{1-\mathcal{R}}}{1+\frac{T}{\tau}\frac{\mathcal{R}}{1-\mathcal{R}}} = \frac{\frac{T}{\tau}\mathcal{R}}{1-\mathcal{R}+\frac{T}{\tau}\mathcal{R}} = \frac{\frac{T}{\tau}\mathcal{R}}{1-\left(1-\frac{T}{\tau}\right)\mathcal{R}}$$

Equivalent to system we previously developed:

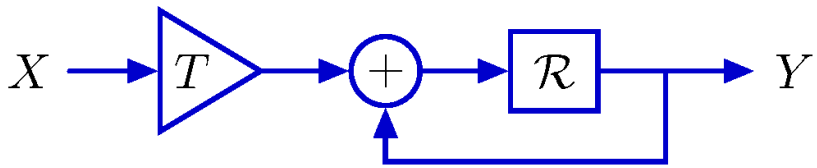
$$y_d[n+1] - \left(1-\frac{T}{\tau}\right) y_d[n] = \frac{T}{\tau} x_d[n]$$

# Model of Forward Euler Method

Replace every integrator in the CT system




with the forward Euler model:




Substitute the DT operator for $\mathcal{A}$:

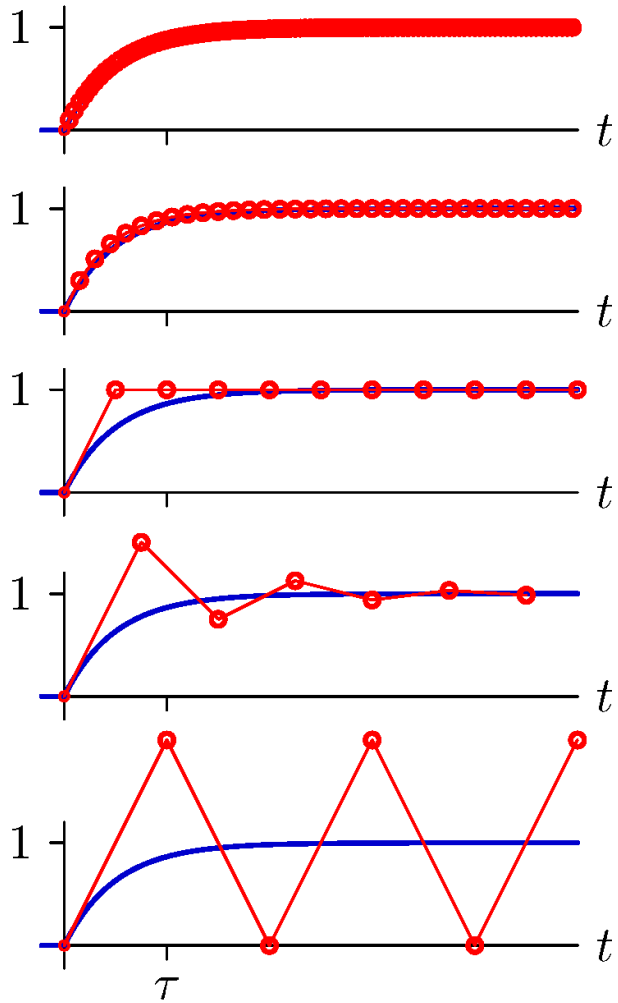
$$\mathcal{A} = \frac{1}{s} \rightarrow \frac{T\mathcal{R}}{1-\mathcal{R}} = \frac{\frac{T}{z}}{1-\frac{1}{z}} = \frac{T}{z-1}$$

Forward Euler maps $s \rightarrow \dfrac{z-1}{T}$.

Or equivalently: $z = 1 + sT$.

# Dependence of DT pole on Stepsize

Pole at $z = 1 - \frac{T}{\tau} = 1 + sT$.



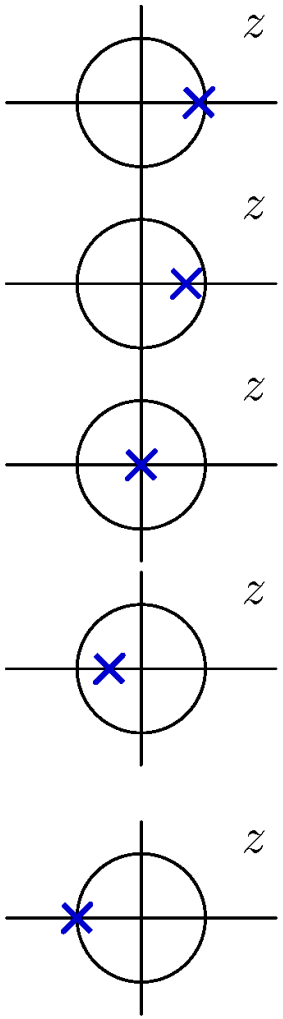

$\frac{T}{\tau} = 0.1$

$\frac{T}{\tau} = 0.3$

$\frac{T}{\tau} = 1$

$\frac{T}{\tau} = 1.5$

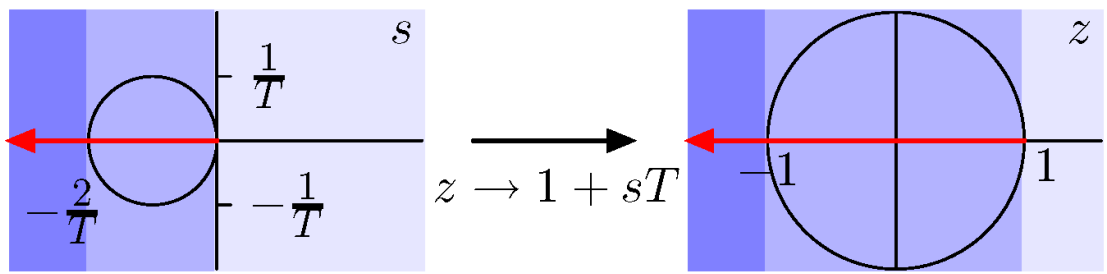
$\frac{T}{\tau} = 2$



# Forward Euler: Mapping CT poles to DT poles

Forward Euler Map:

$$
\begin{array}{ccc}
s & \rightarrow & z = 1 + sT \\
0 & & 1 \\
-\frac{1}{T} & & 0 \\
-\frac{2}{T} & & -1
\end{array}
$$




DT stability: CT pole must be inside circle of radius $\frac{1}{T}$ at $s = -\frac{1}{T}$.

$$-\frac{2}{T} < -\frac{1}{\tau} < 0 \quad \rightarrow \quad \frac{T}{\tau} < 2$$

# Backward Euler Approximation

We can do a similar analysis of the backward Euler method.

Substitute

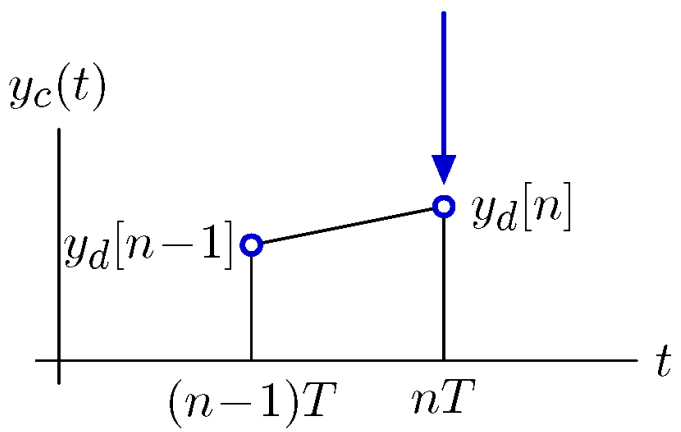
$$x_d[n] = x_c(nT)$$

$$y_d[n] = y_c(nT)$$

$$\dot{y}_c(nT) \approx \frac{y_c\big(nT\big) - y_c\big((n-1)T\big)}{T} = \frac{y_d[n] - y_d[n-1]}{T}$$

$$y_c(nT) = y_d[n]$$

$$\dot{y}_c(nT) = \frac{y_d[n] - y_d[n-1]}{T}$$



# Backward Euler Approximation

We can do a similar analysis of the backward Euler method.

Substitute

$$x_d[n] = x_c(nT)$$
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$$\dot{y}_c(nT) \approx \frac{y_c\big(nT\big) - y_c\big((n-1)T\big)}{T} = \frac{y_d[n] - y_d[n-1]}{T}$$

into the differential equation

$$\tau \dot{y}_c(t) = x_c(t) - y_c(t)$$

to obtain

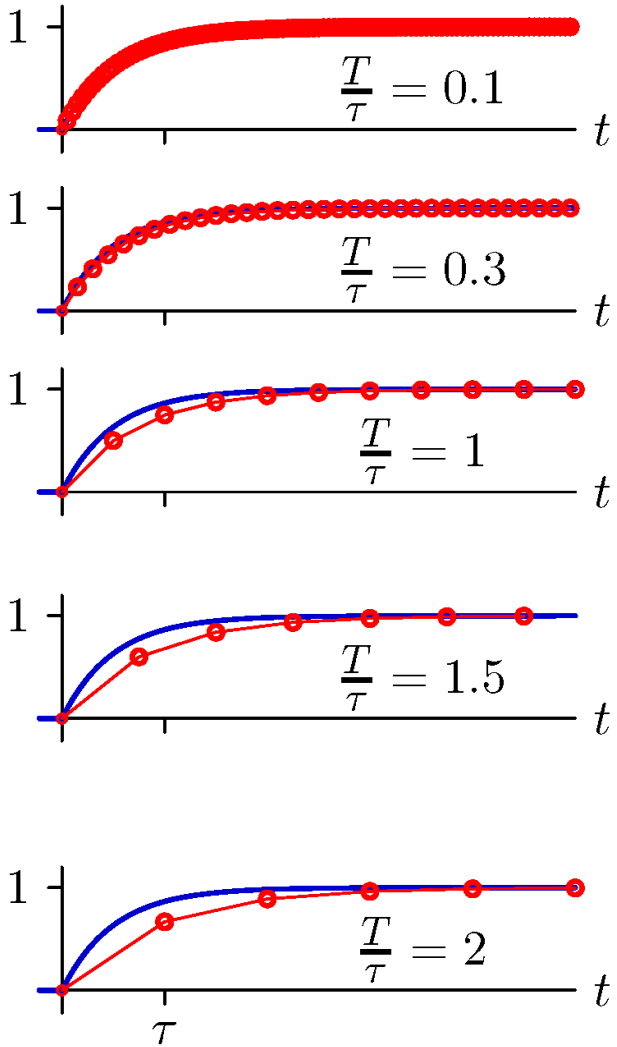
$$\frac{\tau}{T}\Big(y_d[n] - y_d[n-1]\Big) = x_d[n] - y_d[n]\,.$$

Solve:

$$\left(1 + \frac{T}{\tau}\right) y_d[n] - y_d[n-1] = \frac{T}{\tau} x_d[n]$$

# Backward Euler Approximation

Plot.




This approximation is better behaved. Why?

# Check Yourself

DT approximation:

$$\left(1+\frac{T}{\tau}\right) y_d[n] - y_d[n-1] = \frac{T}{\tau} x_d[n]$$

Find the DT pole.

1. $z = \frac{T}{\tau}$
2. $z = 1 - \frac{T}{\tau}$
3. $z = \frac{\tau}{T}$
4. $z = -\frac{\tau}{T}$
5. $z = \frac{1}{1+\frac{T}{\tau}}$

# Check Yourself

DT approximation:

$$\left(1+\frac{T}{\tau}\right) y_d[n] - y_d[n-1] = \frac{T}{\tau} x_d[n]$$

Take the Z transform:

$$\left(1+\frac{T}{\tau}\right) Y_d(z) - z^{-1} Y_d(z) = \frac{T}{\tau} X_d(z)$$

Find the system function:

$$H(z) = \frac{Y_d(z)}{X_d(z)} = \frac{\frac{T}{\tau} z}{\left(1+\frac{T}{\tau}\right) z - 1}$$

Pole at $z = \dfrac{1}{1+\frac{T}{\tau}}$.

# Check Yourself

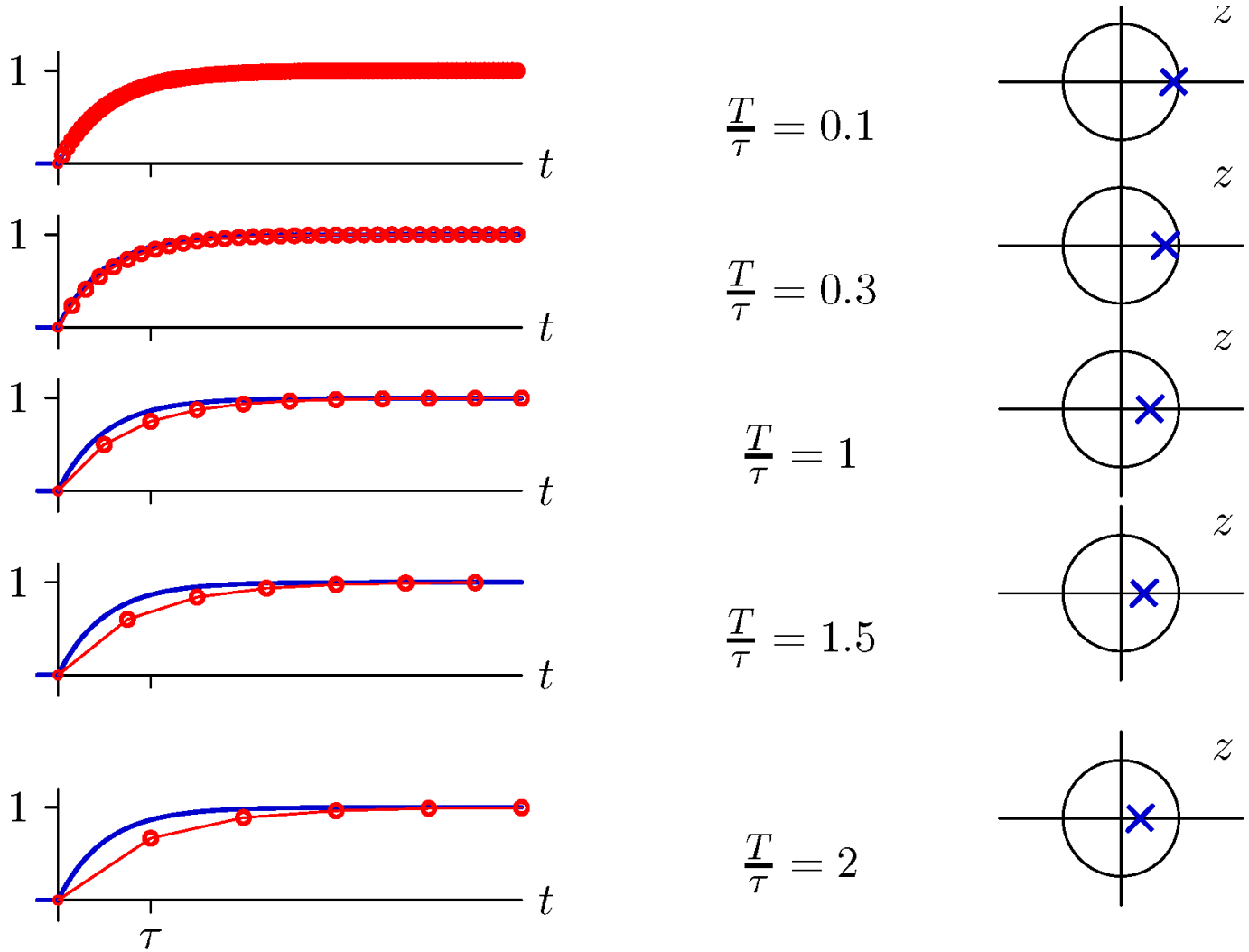
DT approximation:

$$y_d[n+1] - \left(1 - \frac{T}{\tau}\right) y_d[n] = \frac{T}{\tau} x_d[n]$$

Find the DT pole. 5

1. $z = \frac{T}{\tau}$
2. $z = 1 - \frac{T}{\tau}$
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4. $z = -\frac{\tau}{T}$
5. $z = \frac{1}{1 + \frac{T}{\tau}}$

# Dependence of DT pole on Stepsize

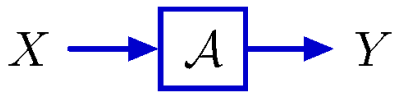



Why is this approximation better behaved?

# Dependence of DT pole on Stepsize

Make a systems model of backward Euler method.

CT block diagrams: adders, gains, and integrators:

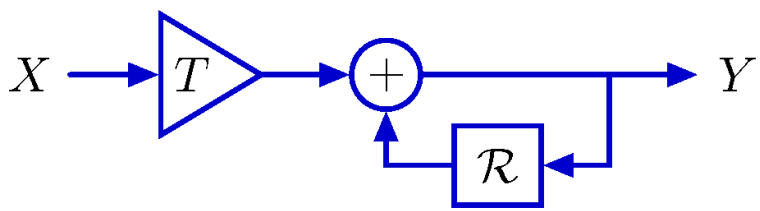



$$\dot{y}(t) = x(t)$$

Backward Euler approximation:

$$\frac{y[n] - y[n-1]}{T} = x[n]$$

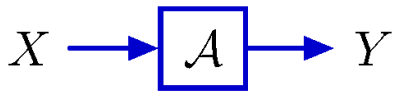
Equivalent system:



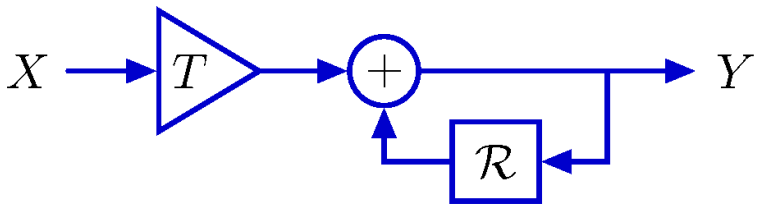

Backward Euler: substitute equivalent system for all integrators.

# Model of Backward Euler Method

Replace every integrator in the CT system




with the backward Euler model:



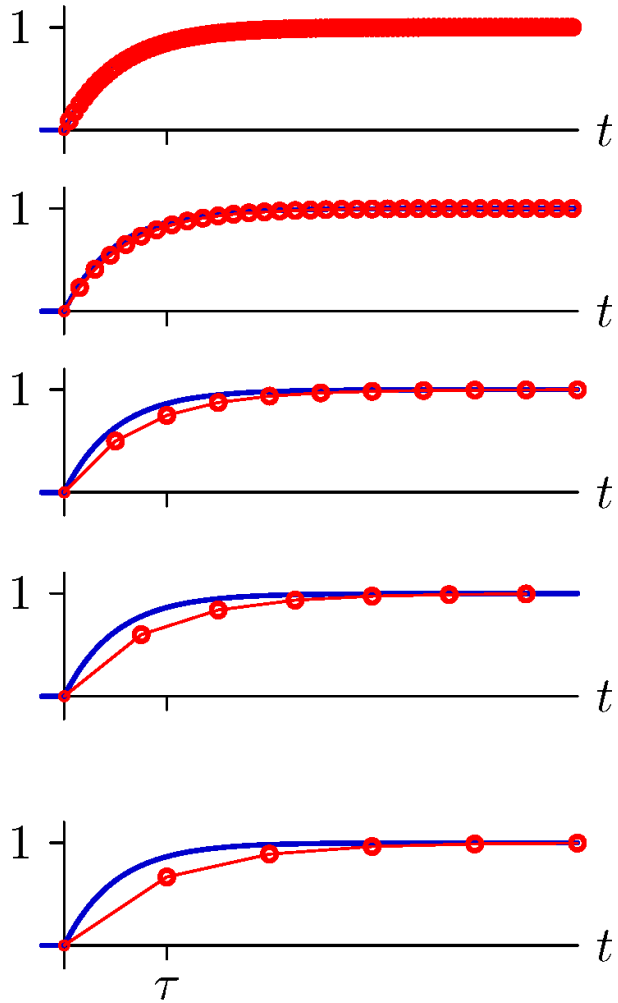

Substitute the DT operator for $\mathcal{A}$:

$$\mathcal{A} = \frac{1}{s} \rightarrow \frac{T}{1-\mathcal{R}} = \frac{T}{1-\frac{1}{z}}$$

Backward Euler maps $z \rightarrow \dfrac{1}{1-sT}$.

# Dependence of DT pole on Stepsize

Pole at $z = \frac{1}{1+\frac{T}{\tau}} = \frac{1}{1-sT}$.



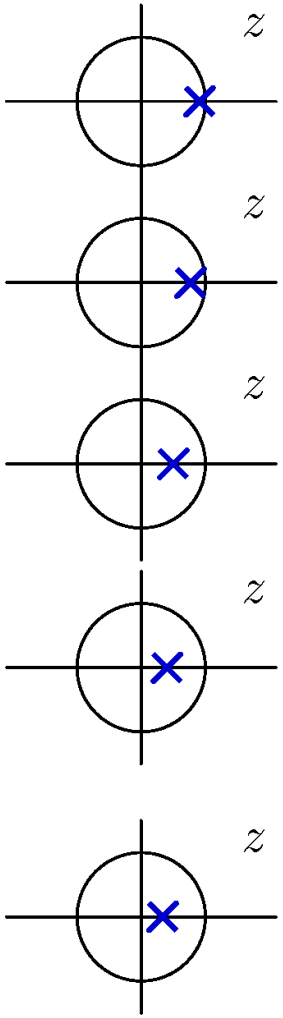

$\frac{T}{\tau} = 0.1$

$\frac{T}{\tau} = 0.3$

$\frac{T}{\tau} = 1$

$\frac{T}{\tau} = 1.5$

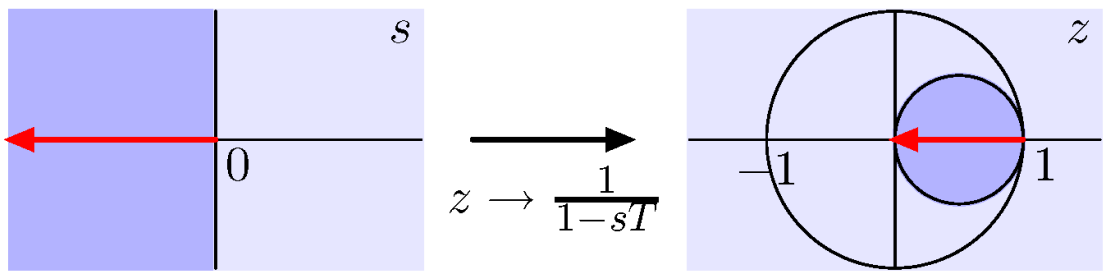
$\frac{T}{\tau} = 2$



# Backward Euler: Mapping CT poles to DT poles

Backward Euler Map:

| $s$ | $\rightarrow$ | $z = \frac{1}{1-sT}$ |
|---|---|---|
| $0$ | | $1$ |
| $-\frac{1}{T}$ | | $\frac{1}{2}$ |
| $-\frac{2}{T}$ | | $\frac{1}{3}$ |



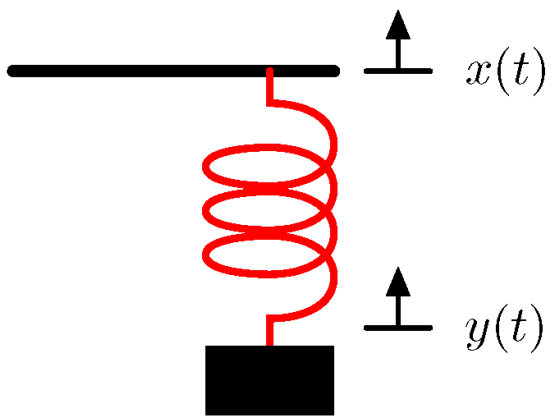

The entire left half-plane maps inside a circle with radius $\frac{1}{2}$ at $z = \frac{1}{2}$.

If CT system is stable, then DT system is also stable.

# Masses and Springs, Forwards and Backwards

In Homework 2, you investigated three numerical approximations to a mass and spring system:

- forward Euler
- backward Euler
- centered method



# Trapezoidal Rule

The trapezoidal rule uses centered differences.

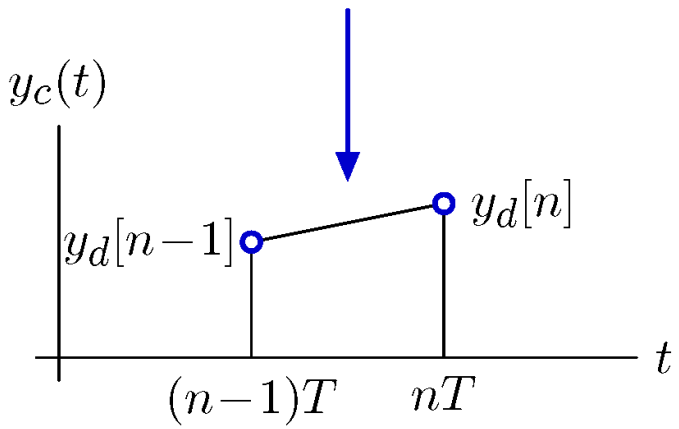
$$\dot{y}(t) = x(t)$$

Trapezoidal rule:

$$\frac{y[n] - y[n-1]}{T} = \frac{x[n] + x[n-1]}{2}$$

$$y_c\Big((n+\tfrac{1}{2})T\Big) = \frac{y_d[n] + y_d[n-1]}{2}$$

$$\dot{y}_c\Big((n+\tfrac{1}{2})T\Big) = \frac{y_d[n] - y_d[n-1]}{T}$$



# Trapezoidal Rule

The trapezoidal rule uses centered differences.

$$\dot{y}(t) = x(t)$$

Trapezoidal rule:

$$\frac{y[n] - y[n-1]}{T} = \frac{x[n] + x[n-1]}{2}$$

Z transform:

$$H(z) = \frac{Y(s)}{X(s)} = \frac{T}{2}\left(\frac{1+z^{-1}}{1-z^{-1}}\right) = \frac{T}{2}\left(\frac{z+1}{z-1}\right)$$

Map:

$$\mathcal{A} = \frac{1}{s} \rightarrow \frac{T}{2}\left(\frac{z+1}{z-1}\right)$$

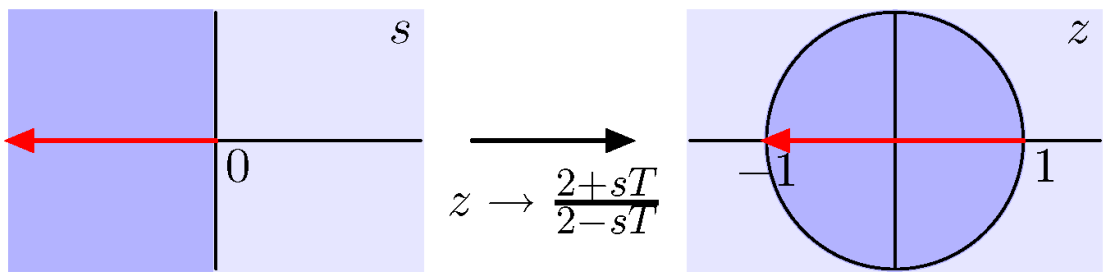
Trapezoidal rule maps $z \rightarrow \dfrac{1+\frac{sT}{2}}{1-\frac{sT}{2}}$.

# Trapezoidal Rule: Mapping CT poles to DT poles

Trapezoidal Map:

$$s \quad \rightarrow \quad z = \frac{1+\frac{sT}{2}}{1-\frac{sT}{2}}$$

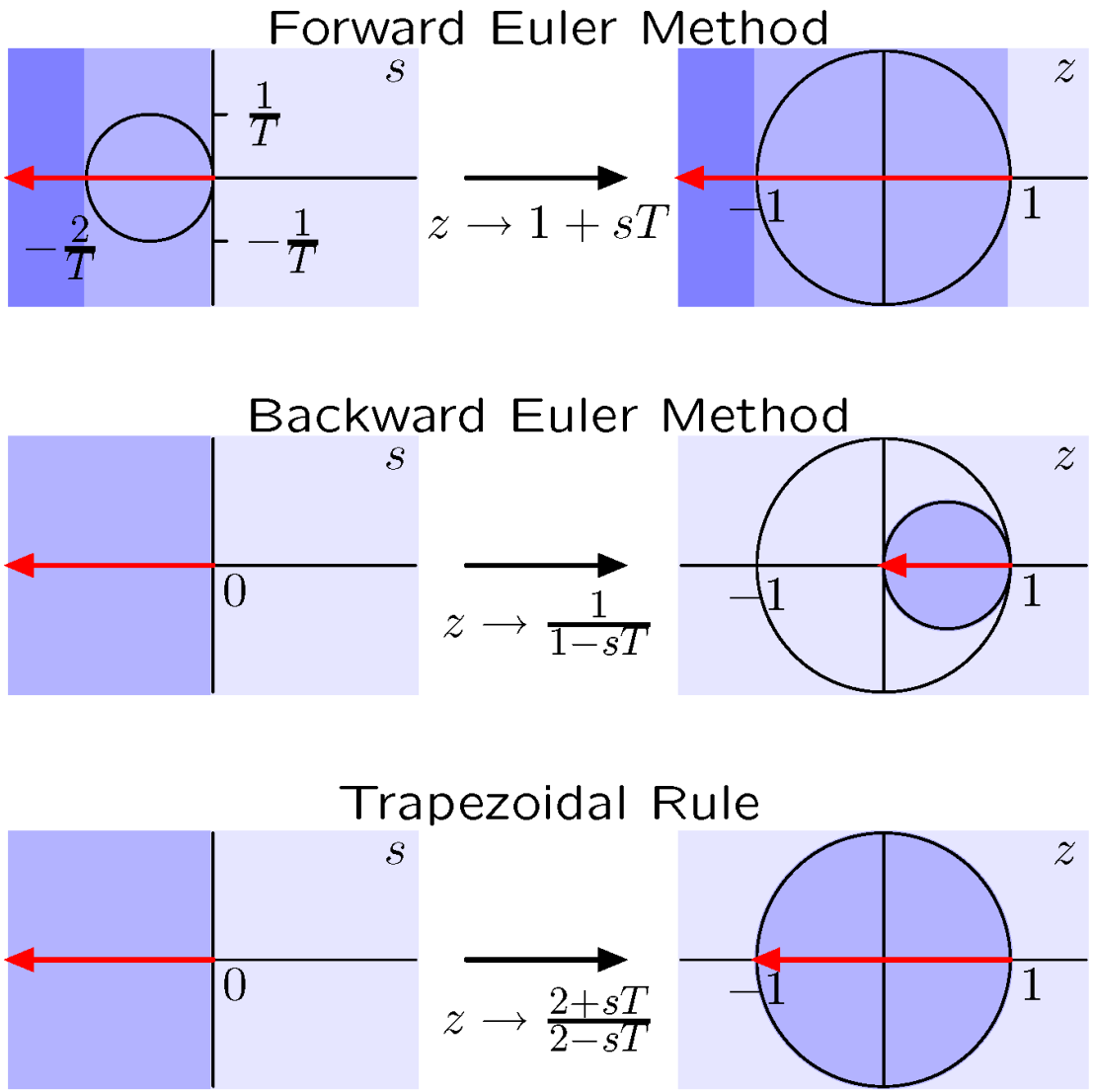
| $s$ | $z$ |
|---|---|
| $0$ | $1$ |
| $-\frac{1}{T}$ | $\frac{1}{3}$ |
| $-\frac{2}{T}$ | $0$ |
| $-\infty$ | $-1$ |
| $j\omega$ | $\frac{2+j\omega T}{2-j\omega T}$ |



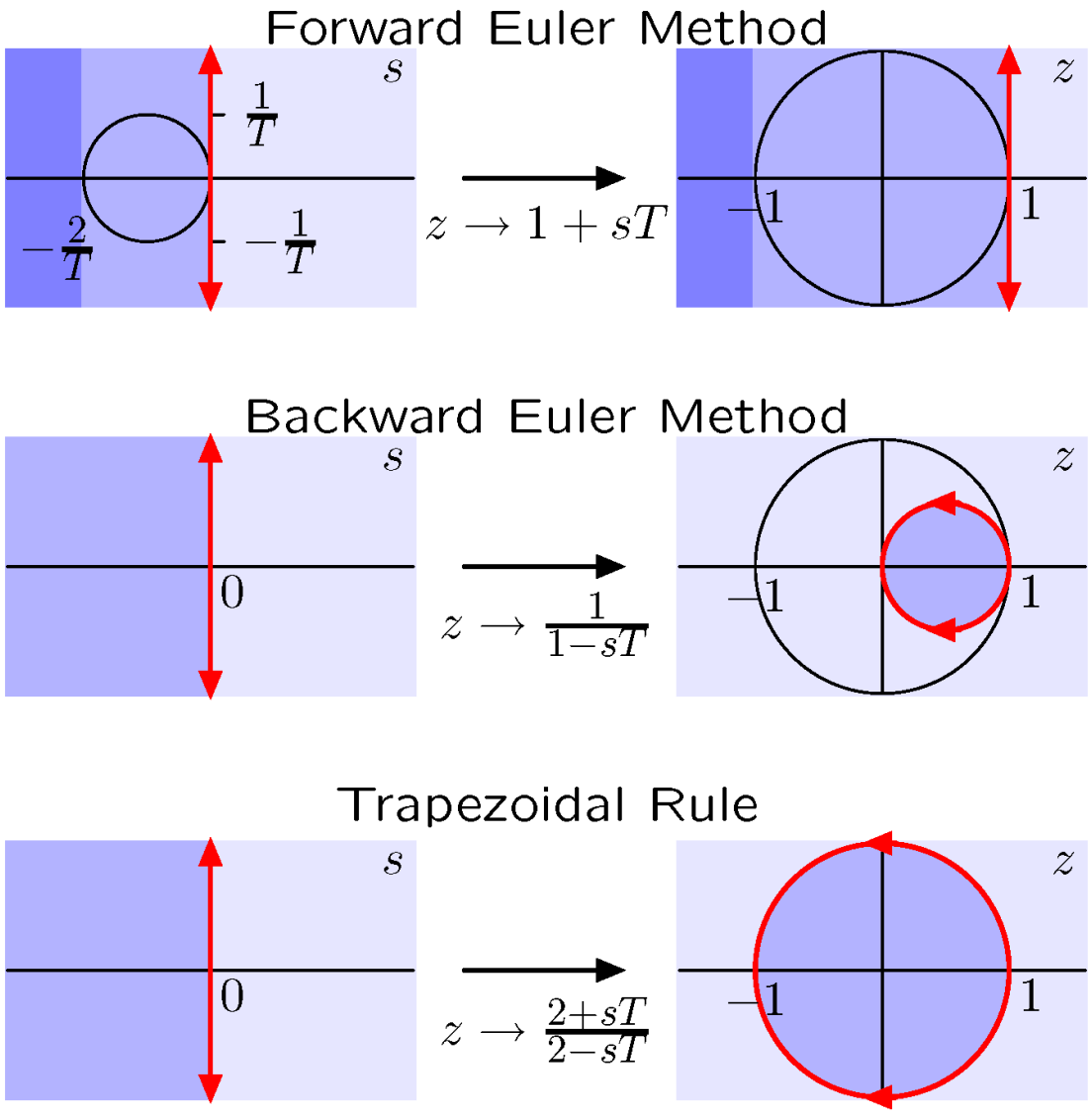

The entire left-half plane maps inside the unit circle.

The $j\omega$ axis maps onto the unit circle

# Mapping s to z: Leaky-Tank System

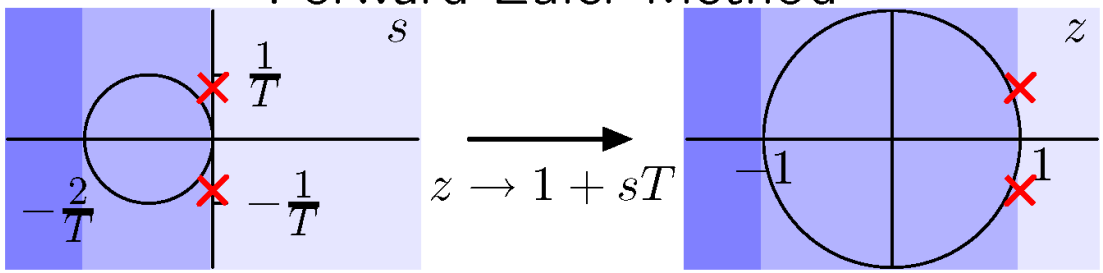


# Mapping s to z: Leaky-Tank System

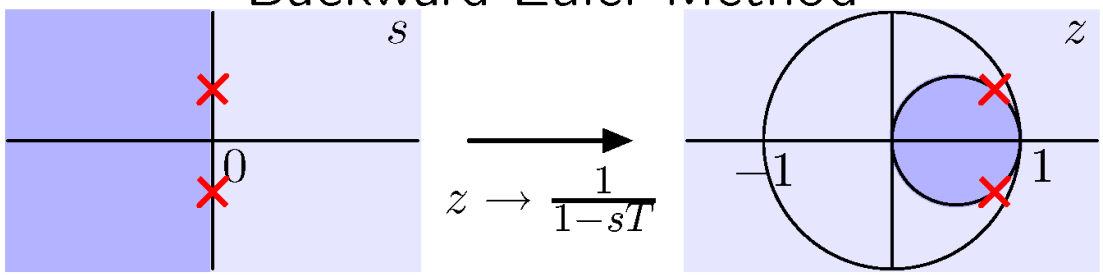


# Mapping s to z: Leaky-Tank System

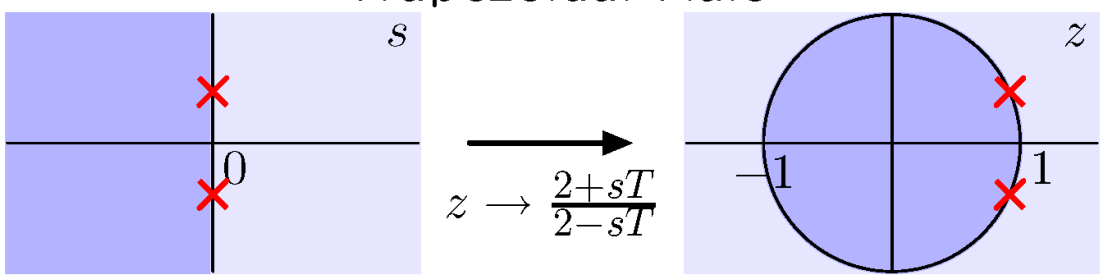
Forward Euler Method




Backward Euler Method




Trapezoidal Rule



# Assignments

- Reading Assignment: Chap. 10.1-10.3, 10.5-10.8
- Homework 2: Due by Mar. 17, 2025